

Group 14 : Phase 1

01

Volunteers in Animal-Shelters and Foster-care

The Team



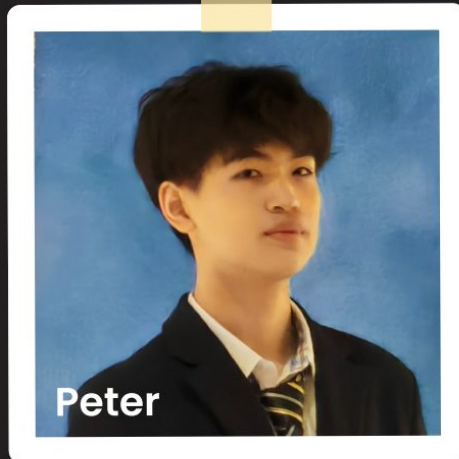
CCO

Using graphical communication to clearly convey goals and outcomes.



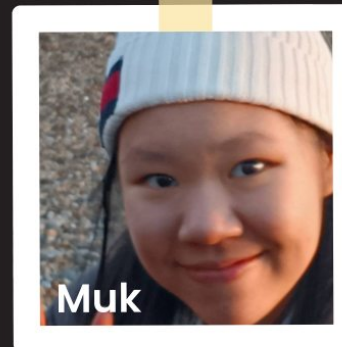
COO

Overall project management: tracking & planning.



CIO

Gather and analyse all the information.



CTO

Technical analysis of benchmark products and research data.

1 Defining the Underserved Group

In the United Kingdom, over **250,000** companion animals (cats and dogs) are **surrendered** annually and either enter the foster-care or shelter system, often having experienced trauma, disrupted attachment and unstable living conditions. [1]

Rising abandonment rates, driven largely by economic pressures, have placed significant **strain on animal welfare organisations**, many of which operate beyond capacity. Faced with such a difficult resource allocation problem, some shelters consider the euthanasia of unadoptable animals to allow for new intake.



51% [2]

increase in **animal abandonment** cases in the UK (2021-2024)



24% [3]

increase in the **surrendering** of pets due to the cost-of-living crisis in 2022

Core Animal Care Responsibilities in Shelters



Intensive cleaning of kennels is done daily to maintain hygiene and prevent the spread of disease.



Distributing appropriate food and fresh water to all animals, following specific dietary needs.



Grooming animals to keep them comfortable and presentable.



Health monitoring animals for signs of illness. Administering medications such as pills, vaccines or specialised treatments for injured animals.

Shelter Statistics

≈5-7 days [4]

the average length of stay in shelters before adoption, rescue or euthanasia

≈7000 [5]

healthy dogs are **euthanised** daily in the UK

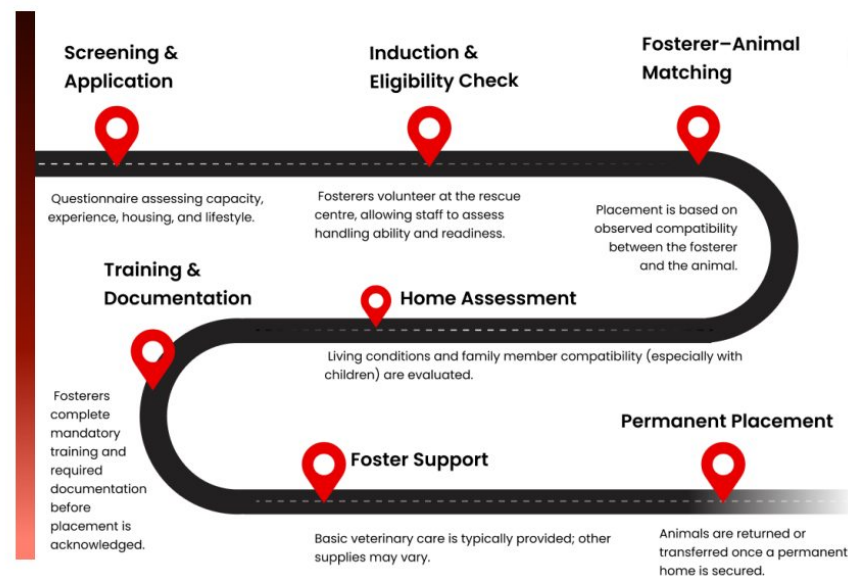
≈£500,000 [6]

required to operate a medium-size shelter (60 dog) annually

2 The Significance of Fostering

Fostering **reduces the strain** faced by **freeing up a shelter's capacity** to rescue more animals by placing them into a home environment. This also supports the **rehabilitation** of animals that may have experienced neglect or trauma exhibited through **behaviours issues** (#1 cause of relinquishment) that makes them **undesirable as pets**. [7]

Fosterer's Timeline



“The shelter's in UK are **very strict** about who they want in-contact and fostering their animals. They even **reject some applicants with young kids or no gardens.**”
-Dominique (Fosterer)

Initial research shows that the fostering process in the **UK** is **time and resource intensive**, with **strict screening** that can lead to **over-rejection** of potential volunteers. Applicants must **provide time, space and personal financial resources** to care for the animal, usually without financial incentives.

Project Direction

HMW **reduce pressure on foster-care volunteers and organisations**, improve rehoming outcomes and **support the creation of stable environments** for animals transitioning into permanent homes.

User Group Rationale

Purpose

To understand the significance of the user group

04

1 Pathways Through the Animal Shelter System

1.1 Origin / Relinquishment

Animals are most commonly relinquished due to **behavioural issues**, followed by **housing changes or relocation**, and an inability to provide care, often due to **financial constraints, limited time, or illness**.

Potential user groups involved:

Pet owners, Animal welfare Organisations, Informal care providers

1.2 Intake, Assessment & Stabilisation

An animal shelter provides **temporary housing, food, health care, and socialisation for surrendered pets**. Here, they are screened based on **physical well being** and **temperament** to identify suitability for societal reintegration. They are **vaccinated** and **microchipped** where appropriate.

Potential user groups involved:

Animal welfare Organisations, Animal shelters, Animal rescue centres, Vets, Volunteers

Rehoming OUTCOMES

Scenarios

Adoption

Humane Euthanasias

Fostering

Rescue Centers

We have decided to research **deeper into fostering and rescue centres**, as the people working in these environments are often an underserved group whose daily challenges and labour receive far **less attention than adoption or humane euthanasia outcomes**.

2 Why are Volunteers Underserved?

Temporary-care such as rescue shelters and fostering is a critical stage in animal rehoming, but it can be challenging due to **limited resources** and **no financial incentives**. In turn, volunteers continue out of passion to save lives, despite the **work often being thankless**.



“

“I don’t do it for fun, I do it to save lives”

Fatima (Fosterer)

“it can be emotional draining
...the work never really stops”

Interviewed Shelter Staff



Unlike traditional long-term pet ownership, shelters face **high animal intake** and rapid turnover rates. Volunteers and shelter staff help create a stable environment through **routine but demanding tasks such as cleaning, feeding, waste handling and health monitoring**. These essential responsibilities support animal welfare and shelter operations, yet the **people performing them are often under-recognised**.

Therefore, the following research explores product opportunities and solutions to **support and alleviate the burden of the volunteers and organisations caring for animals** behind the scenes to give them the **best chance of thriving** in their forever home.

2

Ethical Consideration

We **interviewed volunteers and shelter staff** because they are directly involved in temporary animal care and can authentically describe its **demands and constraints**.

3.2 Why Volunteers?

We acknowledge that **working directly with animals can be unethical**, especially in user testing given their often sensitive rescue backgrounds. As such, we decided to avoid working directly with animals but instead with the individuals/ volunteers that create a stable and clean environment for them.

Interviews and observations provided first-hand insights into caregiving practices while **avoiding stress to the animals**, supporting a more responsible design process.

3.3 Future: Idea Assessments

Future prototypes should **avoid direct interaction with animals**, as user testing involving animals raises ethical concerns and often requires complex approval processes. Focusing on **human caregivers instead enables responsible development** while still **addressing the animals’ needs**.



Ethical Practices

Names of individuals have been anonymised in accordance with ethical guidelines from Imperial College Interview participants were given a consent form and verbally informed of their rights.

Research Planning

Purpose To **plan** and **summarise** research operations

05

1 Research Approach

Our research focused on identifying **tangible caretaker problems** narrowed down to **domestic animals: cats & dogs**, as they are the most common in household.

- Context and Scale
- Operations and Resources (daily labour, tools, budget)
- Volunteers and Training
- Animal Welfare and Behaviour
- Future Needs and Innovation

These research areas were **translated into interview questions** and guided our subsequent research.

Research Work-flow



2 Interviews

x3 Animal Shelters

Rescue centres are non-profit organisations that **rely heavily on volunteers** and often operate at high capacity due to more animal surrenders than adoptions. These interviews **aims to identify physically demanding, repetitive, or difficult tasks**, which is important because high volunteer turnover leads to varying experience and skill levels.



8 Interviews (15-60min)

Across 4 countries

+35 Key Insights Identified

Over 2000 re homed

x2 Independent Fosterers

Fosterers provide **temporary homes** for rescued animals. We interviewed two fosterers from different countries with different mindsets to **identify the challenges of caring for multiple animals alone**. As solo caretakers, they face greater **physical and financial strain**, often without organisational support.

Fatima (UAE) Experienced fosterer



+100 Cats Rehomed

Prioritise quantity: Care for as many cats as possible using the same pool of resources.

Dominique (UK) Dog Fanatic



14 dogs owned

+70 years old

Prioritise quality: Focus on fewer animals, including traumatised dogs with behavioural issues.

x3 Pet Owners

We **benchmarked** the experiences of rescue centres and fosterers against those of pet owners who acquired their animals through non-shelter routes (strays & private sellers.) This comparison aims to **identify differences in caregiving practices, expectations and challenges** across both pathways.

Owners:



12 Cats
(Strays & Home Birth)



1 Dog
(+3 caretakers)



1 Dog & 2 Stray Cat
(+3 caretakers)

3 Observations: Java Whiskers Cat Cafe

Fostering Home and Cat Cafe



Goal: Understanding the **practices** that goes into **creating a successful environment** for fostering.

Hygiene:

- All cats groomed with minimal shedding on furniture
- Potty trained to enter a separate area to pee/poo, keeping the environment clean, and odourless
- Guests are advised to wash hands and remove shoes before entering area with cats → strict hygiene standards

Feeding:

- Simple set up with cats trained to just take what they need

Cleaning:

- Manual broom and dustpan

Atmosphere:

- Sound insulated, low volume zone
- constant whale like sound/music makes a soothing/calming atmosphere

3.2 Pet Owner Survey

30+ people

Goal: Understand whether owners can realistically keep up with the responsibilities of raising a pet.



65% Vaccinated
78% microchipped
87% neutered



80% Vaccinated
87% microchipped
68% neutered

Key Insights

Purpose

To **extract** and **group insights** to **identify opportunity areas**.

06

1 Grouped Insights

1 Health Monitoring & Prevention

Cats often **hide signs of illness** and withdraw when unwell, meaning **only experienced caretakers can recognise subtle abnormalities** (e.g., reduced drinking, wet food preference, frequent toilet use, hot ears). This makes care-taking challenging and **unsuitable for beginners** and casual volunteers.

Young kittens have **higher mortality risk** because they cannot be fully **vaccinated until around six months** of age. During this vulnerable period, they are more susceptible to disease and may also **transmit infections** to other cats in the same environment.

The risk of **cross-infection** remains persistent (notably **respiratory** and **faecal-oral** transmission pathways), and **airborne spread** is particularly difficult to prevent comprehensively.

Community feeding is fragmented across many different sites, which makes **coordination, monitoring, and overall management** much more difficult.

2 Waste-Management

Faeces, expired kibble and fur can be composed alongside enzymes and food-waste to **produce compost**. However, faeces must be air-dried first since they release gases that can kill plants.

Some people have an **innate psychological aversion** to cat waste, and because **standards of "clean"** vary from person to person, volunteer cleaning work often needs to be **redone** due to inconsistent results.

Dogs usually signal when they need the toilet and go outside, **reducing indoor smell but requiring owners to be present**.

Cat waste management **requires regular attention** and is **largely manual**, often generating significant household waste.

Some owners manage fur in their dogs by giving them a **blow out outdoors** to control the amount of loose hair that comes off inside the house

Origin: Animal Shelters Fosterers Pet-owners

3 Cleaning & Grooming

Standard **vacuums devices are often ineffective** as pet fur and fluids cling to furniture and surfaces, promoting bacterial growth. So, animal-safe solutions such as white vinegar and specialised cleaning sprays are used.

There is an opportunity for a inclusive **all-in-one tool** that simplify the grooming and clean-up process simultaneously.

Existing tools (e.g. vacuums) are **typically noisy** and single-purpose, making them **unsuitable for confined cage environments** and for **cats** that are highly stress-sensitive.

Cleaning and disinfection account for the **majority** of the daily workload, including routine sanitisation of cages, glass doors, food and water bowls, and floor surfaces.

Dogs require frequent showers (every few days) due to hormones that causes bodily odor. This takes up a substantial amount of time for owners and is often not enjoyable for the dog

“

“When they’re sick, they **give you signs**. For example, if you offer them wet and hard food and they only eat the wet, they might be feeling teeth or gum pain.”

“I had a lot of beautiful kittens **died from outbreaks** due to them not being underweight and not vaccinated in time”

“At this stage, even though we’ve already implemented one cat per cage, what I still can’t **prevent is cross-infection of respiratory illnesses.**”

“My caretaking method is all very natural, the **poop goes back into compost as fertiliser**, but you got to dry them first or else the gases will kill the trees”

“Volunteers do help me, but they still **don’t meet my hygiene standards**...lots of people still find it hard to deal with waste, even with gloves on.”

“We usually pick up the poo with a leaf scraper and toss it with the **garden compost**. If it’s poo during a walk we’ll pick it up with a plastic bag and toss it with the **general waste**” – Owner

“Most of the cats **hate loud sounds and taking baths**. It would be great if there is a tool that can groom fur, dry-clean and sterilise them at once.”

“If I had the right tools, I could **cut my workload by more than half**...you learn to DIY a lot of tools you don’t have”

if the cats have gone outside we use wipes, **dry sprays** and combs to clean them up – we’ll usually know if they need an extra bath **by their smell**”

User Personas

Purpose

To **translate** insights into **user personas** and **journey maps**

07

Animal Shelter Volunteer

Committed Carly

27 years old
Cat Shelter Volunteer

Enjoys being with animals but dislikes cleaning their waste.

Needs:

- **Low-noise** cleaning tools to avoid **stressing** cats in confined environments.
- **Pet-safe, low-odour** solutions that still meet a "high hygiene" standard.
- **Standardised, repeatable** workflow to reduce person-to-person variability and rework.
- **Cross-infection control** features and easy to **disinfect, modular** attachments for different tasks



Independent Fosterer

Dedicated Darcy

52 years old
Experienced Fosterer

Knowledgeable but finds prolonged physical tasks difficult.

Needs:

- **All-in-one product** that simplifies home-cleaning directly at dealing with animal waste that is pet-friendly
- Standardised solution that can **deal with multiple animals** and used as a shared resource
- A method to **automatically flag-up abnormalities** in specific animals amongst a large group.
- **Reduce the workload, time, and physical** labour placed on a **single caretaker**, while maintaining the same standard of care for multiple dogs.



Casual Pet-owner (Benchmark)

Hands-on Hans

20 years old
First-time Owner

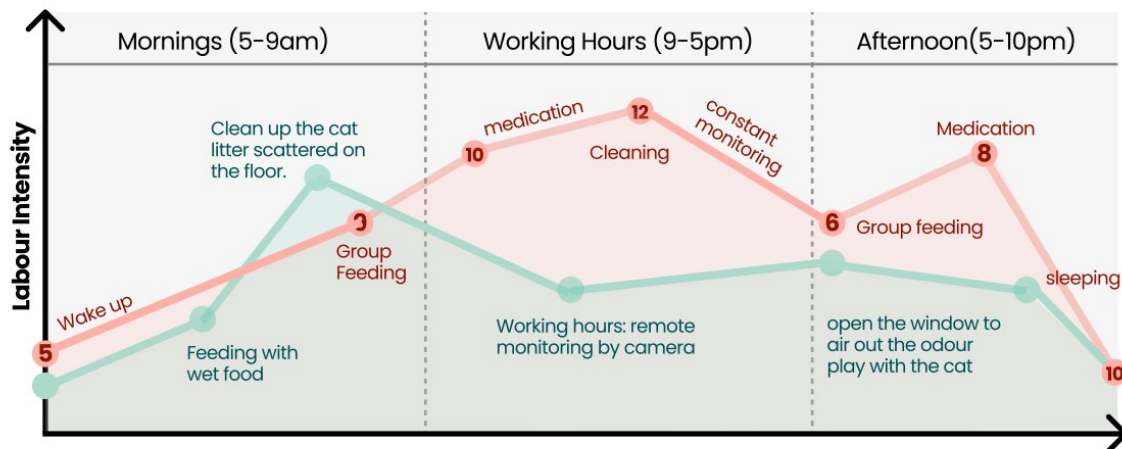
Beginner but wants to become a competent owner.

Needs:

- Device that assists with **odor management of indoor animal waste**, and **reduce the amount of human intervention** required for waste management tasks
- Cleaning technologies that **eliminate bodily odor** of pets, and **residue on their fur without** requiring a **full bath**
- Devices that emit **low noise levels** (especially for cats), and **doesn't cause a sense of confinement** for the animal
- Alternative solutions to **reduce household waste** generated by **waste from pets**



2 Journey Map: Darcy (temporary-care) VS Hans (permanent-care)



Pain-points

- Fostering can become almost a full-time job due to manual monitoring of many animals.
- **Cleaning is the most labour-intensive** task in both scenarios.
- **Medicating** is more labour-intensive than feeding because it must be **done individually**.

3 Barriers to Fostering and Limitations of Group Care

Fostering **through registered organisations** has a **higher barrier to entry** compared to independently fostering strays, due to strict regulations and lengthy approval processes. In group-care settings, attention is divided, making individualised care difficult and resulting in more standardised management. **Therefore, group solution should be scalable and effective across multiple dogs rather than tailored to a single animal.**

3.2 The Commitment to Fostering

Most **care taking responsibility** often falls on **one person** in a **home environment**, despite shelters emphasising that "**fostering should be a whole family commitment, not just an individual's responsibility.**" - Dominique (fosterer)

Thus, there is a demand for devices that increase **caregivers' flexibility, reduce time demands** and **reward users for their time.**



"It's hard work, my grown-up kids doesn't help so **I do everything myself...**"

Market Research

Purpose

Analysis of **existing pet cleaning products** and **identifying gaps** in the market for large scale cleaning

08

1 Market Size and Demand



Target User:

Volunteers and workers in dog and cat shelters and foster-care

Requirements:

Durable, hygienic, cost saving, and new user friendly

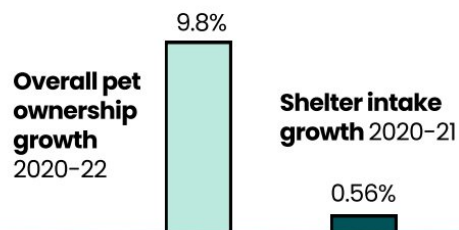
4.8 million cats and dogs intake into **U.S. shelters** in 2024

\$5.5 billion global pet cleaning product market in 2024 with a projected **7.8% CAGR** until **2031** [8]

Due to the 'Pandemic Pet' boom during Covid, in the **US pet ownership growth increased 9.8%** from 2020–2022 (compared to 4.7% between 2010–2019)

On the contrary, **intake levels at U.S. shelters had increased by only 0.56%** from 2020 to 2021. Market growth and development is therefore concentrated in the household level rather than shelters. [9]

Pet ownership/intake growth in U.S.



2 Existing Products

Skorter | Low-Noise Grooming Vacuum with Full Pet Hair Clippers and Brush Kit [10]

£235.99



Pros:

- **Multiple grooming tool heads** using the same body reduces the need for multiple individual tools
- **Low noise** to avoid scaring animals
- Vacuum element embedded with grooming tools ensures a **low mess, high hygiene** set up especially for indoor setups

Cons:

- **Low portability** as design is **bulky**, requires a power outlet, and has no modes for transportation
- Relatively **high cost** compared to individual tools

Handi Scoop PRO Pooper Scooper [11]



Pros:

- Made of **87% Recycled material**
- **Ergonomic handle** with **rotatable scoop jaw**
- Ensures workers or volunteers will not have to come into contact with animal waste directly
- **Reduces bending physical strain**

Cons:

- **High waste solution:** does not solve 1 plastic bag for 1 trip issue
- Relatively **bulky** to carry around
- No place to **store the plastic bag of waste**
- **Visual appearance resembles walking aid** for the less abled



3 Identifying Opportunities

S

- **Large product variety** due to market competition
- **Large general market** due to growth in domestic pet ownership
- **Innovative products** that reduces need for user skill and time labour to facilitate increase in new pet owners following the pandemic

W

- **High cost** especially when products involve automation
- **Targeted at few pets/households** rather than large scale operations
- **High waste** due to concerns with hygiene and lack of incentive
- **Low customisability** as designs generalise for the wider market

O

- Cleaning products that can **operate on a large scale** without having to be **large in size**
- Product with **high reusability** without generating high waste
- Cleaning products that take into account the **relationship dynamic** between **traumatised animals and inexperienced carers**

T

- **Shelters have high hygiene standards** that they are required to upkeep which may limit approaches which adopt a low-waste approach
- Shelters are **extremely diverse** in their **infrastructure** and how they **operate** so products may overgeneralise

Product DFA Analysis Imou Camera

10

Purpose

Analysis of casing design, joining methods, tolerances, repairability, and manufacturing methods

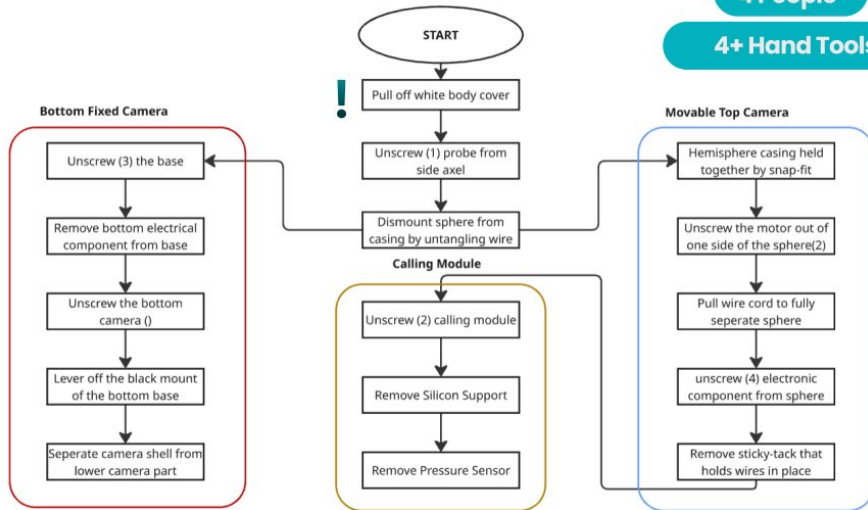
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Decomposition Flowchart

1 hr Disassembly

4 People

4+ Hand Tools



! Main observation: The **minimalist design concealed screws and fixings**, making it difficult to determine **where to begin disassembly**, suggesting it was **not intended to be opened**. Brute force was used to begin dismantling.

2

Manufacturing and Assembly

2.1 Design for manufacturing

Main Manufacturing Process: Thermoplastic Injection Moulding



Uniform wall thickness

Ribbing/ guides

Uniform wall thickness prevents sink marks/warping on the surface of the plastic

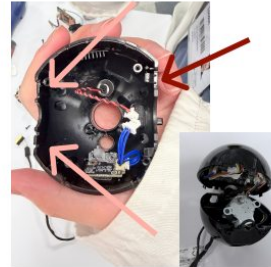
Wall ribbing for structural strength that doubles as assembly guides for snap fits

Non-blended plastics

Consistent, and predictable material behaviour compared to blends or recycled

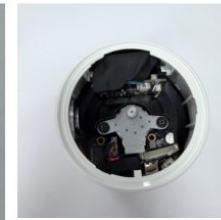
2.1

Part Joining Methods and Pain Points



3 point cantilever snap fit ensures the assembly is tight, but requires 3 contact forces for disassembly which is difficult without using an extra tool as a lever and damaging the plastic surface

A flathead screwdriver was used as a third arm - There could have been more clearance at snap fits to allow parts to be separated more easily



Part Joining Methods and Pain Points

Standardized screws of the same model used throughout the assembly which reduced the need for multiple screw heads, but some screws hidden in between electronic components was difficult to reach

Other notes: Minimal adhesives were used for the assembly, with only tape being used to group up wires for electronic components

3

Repairability

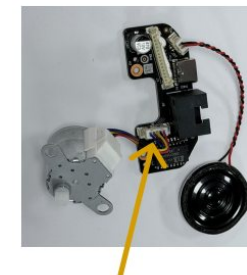
Importance for user group:

Budget limitations in shelters require tools and devices to last as long as possible before repurchasing

Focus:

Electronic components and moving parts as these experience the most wear and are most likely to fail

3 year warranty



Electronic components can be **isolated** and **replaced** as they are joined to the main board using **JST PH connectors**, which are **standard parts** requiring **minimal force** to remove, however, these can only be accessed after disassembling the majority of the main casing, where disassembly is likely to cause **permanent damage**

Project Opportunity

11

Purpose

To **summarise our project** and **explain the focused opportunity** from researches.

1 Summary: Research and Project

This project explores the challenges **temporary-petcare workers** such as **animal shelter volunteers** and **fosterers** face on a day-to-day. Through **contextual research, interviews** (animal-shelters, fosterers and pet-owners across 4 countries) and **market analysis**, our team has identified issues in domains such as grooming, maintaining hygiene and managing pet waste, especially in busy, **resource-limited environments**.

Therefore, in the next phase of the project, our team diverges into the diverse domain of cleaning **in order to better support animal care** and **reduce the resource strain** experienced by animal care-takers.

2 Focused Opportunity: Cleaning

Pet cleaning was selected because it is frequent, labour-and-time consuming, and directly affects animal health and hygiene. Current routines require **unpleasant contact** with **urine, faeces, hair, and contaminated surfaces**, leading to inconsistent standards and higher cross-infection risk, especially under pet-safe cleaning constraints. A **handheld, battery-powered device** can help **reduce direct contact, speed up cleaning and disinfection, and improve consistency**, making animal volunteers and care-takers less resistant to doing these tasks and freeing time for higher-value animal care.



3 Individual Directions



Peter: Multi-tasking

Current pet-care cleaning often requires multiple tools for different tasks, which wastes time and leads to inconsistent hygiene. A modular multi-tasking device with one powered base and interchangeable heads could combine different function in one platform, improving efficiency and user experience.

HMW design a modular handheld pet-care device that combines multiple cleaning tasks into one system to improve efficiency, consistency, hygiene, and safety?



Cici: Waste Management in Shelters

Current pet-care waste handling often requires multiple tools and steps to manage faeces and urine, increasing odour exposure, mess, and inconsistent hygiene.

HMW design dedicated waste-management device could combine waste pickup, sealing, and spot-cleaning in one workflow, reducing direct contact with contaminants while improving efficiency and lowering cross-infection risk.



Preme: Waste Management in Home Environments

Managing pet waste is an unglamorous and repetitive part of pet care. Caregivers constantly pick up faeces, wipe urine and vacuum shed fur to maintain clean environments. Yet once this waste is sealed in plastic and thrown away, its environmental impact is rarely considered.

HMW design an engineerable solution to manage pet waste in multi-dog home environments while reducing labour and resource demands on caregivers?



Muk: Grooming and Maintenance

Grooming in shelters is time-consuming and requires close contact with often stressed or unpredictable animals, creating safety risks for less experienced volunteers.

HMW design a device that makes grooming in animal shelters safer, easier, and more precise for less experienced carers and volunteers?

PetEase



Individual Work by Yuxuan Zhou

CID: 02569052

Introduction & Research

1

Introduction



Current **pet-care cleaning** often requires multiple tools for different tasks, which **wastes time, adds clutter, and leads to inconsistent hygiene**. A **modular multi-tasking** device with one **powered base and interchangeable heads** could combine dry cleaning, grooming, and shed-hair collection in one **platform**, improving **efficiency** and **user experience** while reducing **direct contact** and **cross-infection risk**.

How might we design a modular handheld pet-care device that combines multiple cleaning tasks into one system to improve efficiency, consistency, hygiene, and safety?

2 Similar Existing Products



Dyson Pet Grooming Kit [1]

Oneisall Dog Cat Grooming Kit [2]

Purpose

To Introduce the design and insights

3

Insight



In daily pet care, cleaning is a sequence of small tasks—removing **loose hair**, dry-cleaning **dirt**, and keeping the surrounding area tidy—but these steps are spread across **multiple** tools. Constant switching and setup creates **friction and clutter**, so under time pressure people often skip steps or do a “quick pass,” leading to **inconsistent hygiene** and **repeated re-cleaning**. The barrier is therefore workflow fragmentation and reduced compliance, not a lack of awareness.

4

Insightful Quotes From Interview

“If I had the right tools, I could cut my workload by more than half.”

— HeMaoZhu, rescue centre in China

Ami Cat Rescue Centre in China —

“The vacuum cleaner we bought—well, apart from vacuuming, it doesn’t really have much functionality. And it’s quite loud.”

“Most of the cats hate loud sounds and taking baths. It would be great if there is a tool that can groom fur, dry-clean and sterilise them at once.”

— Independent Fosterer

Opportunities

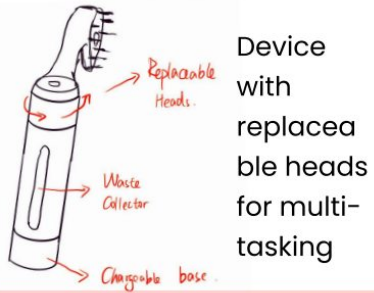
Design a **handheld, battery-charged** platform with a **reusable powered base** and **interchangeable heads** that consolidates key cleaning tasks into one system. This could reduce **switching time** and **storage burden**, improve **consistency** of outcomes, and make routine hygiene easier to complete, while allowing head modules to be washable or replaceable for better contamination control.

Initial Idea Concepts

Purpose

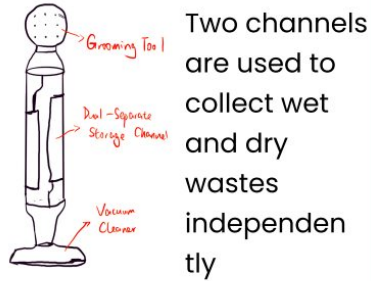
To show 15 Concepts, 3 For Attachments, 3 for storage and 9 combinations

Rotation-Replaceable Heads



Device with replaceable heads for multi-tasking

Dual-Separate Channels



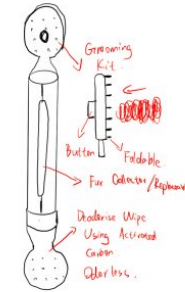
Two channels are used to collect wet and dry wastes independently

Single-Head Vacuum Function



Replaceable Heads of different sizes are used for all conditions of vacuum cleaning, waste is collected at the bottom

Dual-Head Function



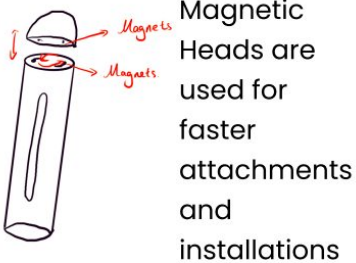
The dual-head design combines grooming and deodorisation by using carbon and can collect fur at the same time for tidiness.

3-in-1 Caring Function



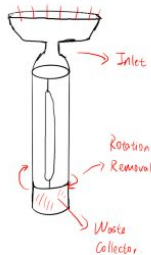
The design combines foldable cleaning features for the whole cleaning process.

Magnetic Snap Heads



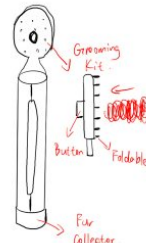
Magnetic Heads are used for faster attachments and installations

One-end rotational Container



The other end is designed to be a container for furs collected from the front tool. Easy to remove.

Single-Head Grooming Function



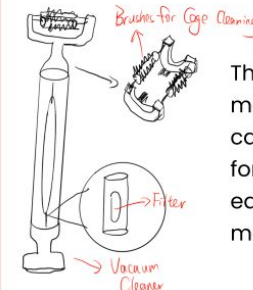
One head is set to do grooming and fur collecting at the same time, while the other end can be removed to clean up the storage.

Surface Tidy Function



This design allows user to tidy up surfaces like desks after grooming using the same tool, saving time.

Shelter Cleaning Combo



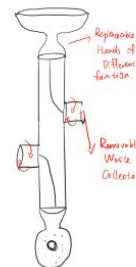
The design makes the cage cleaning for shelters easier and more efficient.

Side Rail Attached Heads



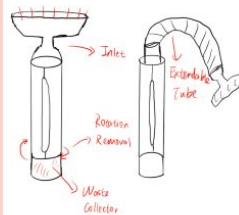
Side Rail Heads are used for easier user experience and better efficiency

Side Channels



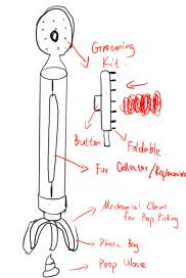
Two side outward storage channels are designed for better storage and removal convenience.

Extendable Single-Head Function



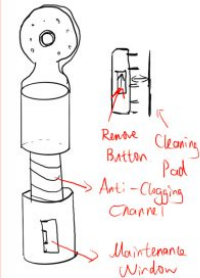
An extendable tube is inserted in the device to help user handle jobs of different angles and distance.

Mechanical and Electrical combination



This design combines electrical and mechanical feature, allowing user to pick up waste and groom at the same time.

Hygiene & Maintenance



This design contains anti-clogging channel and maintenance window for easy repair.

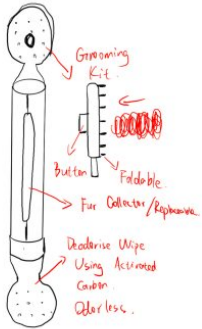
Concept Selection

Purpose

To show the process of selecting the final concept

1

Final Concept



Dual-head care and cleaning system

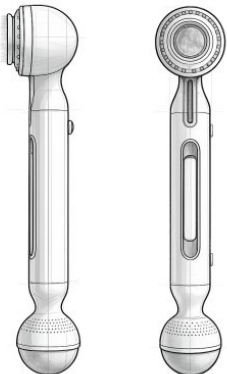
A **handheld dual-head** device that combines complementary **cleaning** and **care** functions in one compact product. It helps shelter staff reduce **tool switching**, remove **loose fur**, freshen surfaces, and improve hygiene more efficiently. The design supports faster daily routines while remaining **lightweight, practical**, and suitable for temporary animal-care environments.

Selection Rationale

- Best balance between **functionality, feasibility, and usability**
- **Easier to understand** than more complex multi-function concepts
- **Lighter** and more **practical** as a handheld device
- More **realistic** to design and develop
- Creates a **clearer** and more **efficient** workflow
- Reduces **tool switching** during shelter care tasks

2

Refined Sketches



Basic Functions

- Exchangeable heads for different tasks
- Grooming and collect the furs by suction, using a fan and motor
- Passive air filter & cleaning via active carbon
- Hygiene Improvement
- Surface cleaning

3

Concept Review

"I like the idea of **combining functions** into one handheld device because in a shelter we do not really have time to pick up different tools for every small task." ----- Shelter Staff

"The dual-head idea makes sense because the functions feel **related**, and that could **save time** during daily care routines. I would be more likely to use something like this" ----- Leo, Independent owner

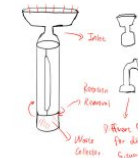
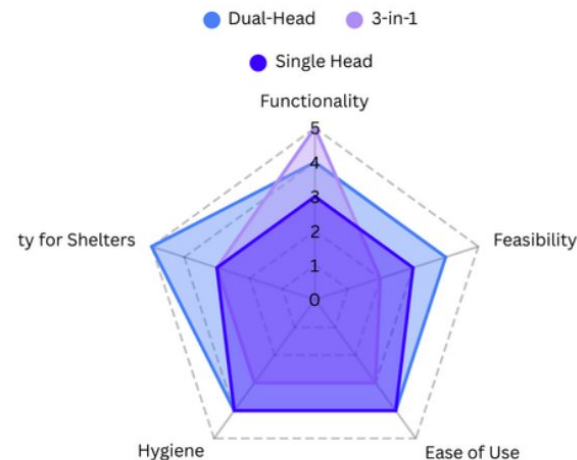
"I like that it seems **compact** and **multi-functional**, because that would make repetitive tasks feel easier. It just needs to stay lightweight and simple to use." ----- Shelter Volunteer

4

Concept Benchmarking

3 concepts are selected from 15 concepts based on user review.

They will be benchmarked according to functionality, feasibility, ease of use, hygiene and compactness.



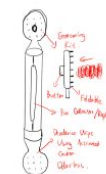
Single-Head Vacuum

Function: Functionality 3, Feasibility 3, Ease of use 4, Hygiene 4, Suitability for shelter use 3.



3-in-1 Caring Function:

Functionality 5, Feasibility 2, Ease of use 3, Hygiene 3, Suitability for shelter use 3.



Dual-Head Function:

Functionality 4, Feasibility 4, Ease of use 4, Hygiene 4, Suitability for shelter use 5.

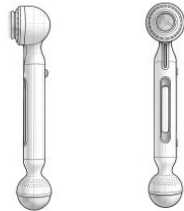
Concept Development

Purpose

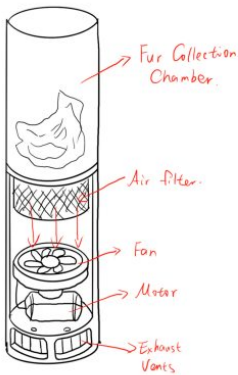
To show the features in final product via sketches and models

1 Product Dimension

- 300 mm length
- 48 mm body diameter
- 62 mm top head width
- 52 mm bottom head diameter

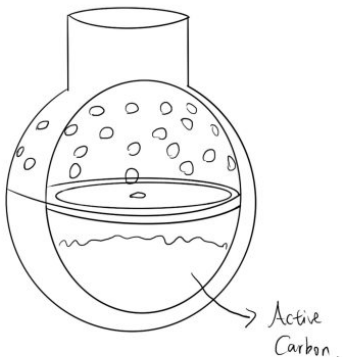


2 Loose Fur Collection



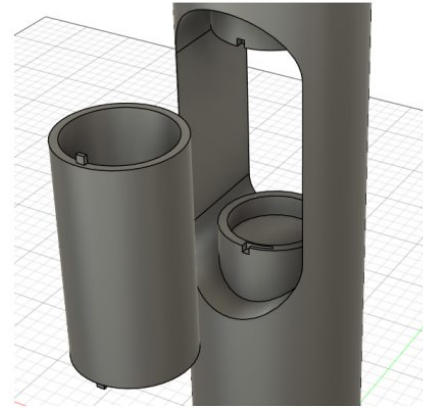
The internal fur collection system is arranged **vertically** within the main body to maintain a compact handheld form. Loose fur is drawn into an upper **collection chamber** positioned above the **motor-driven fan unit**, while a **mesh** or **filter** layer traps hair before it reaches the airflow mechanism. This layout helps reduce **clogging**, improves **maintenance**, and keeps the internal components logically organised.

3 Passive Air Filter



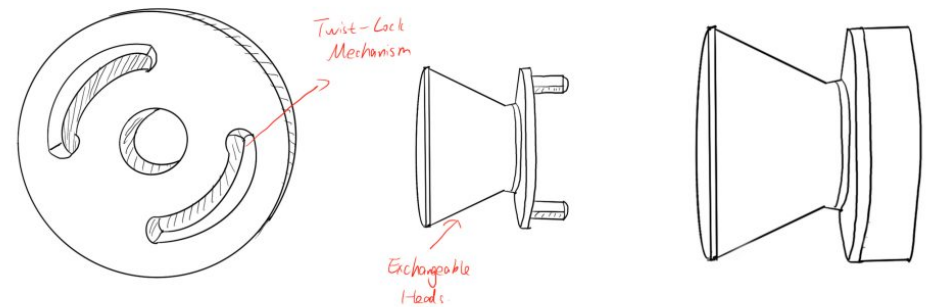
The lower **deodorising** head is designed as a **passive** odour-reduction module with a perforated outer housing and an internal activated carbon cartridge. Air naturally enters through the vent openings and contacts the carbon, allowing odours to be adsorbed without an **additional motor** or **fan**. This keeps the structure **simple, compact, and easy to maintain**.

4 Loose Fur Storage Removal



The fur collection chamber is designed as a **removable cylindrical cartridge** located in the mid-body of the device. It can be accessed through a **side opening**, allowing users to monitor, remove, and clean the chamber easily. A simple **twist-lock** mechanism is used to secure the cartridge in place, keeping the structure **compact, stable, and practical** for repeated use.

5 Head Replacement for multi-tasking



The **interchangeable** upper head uses a simple **twist-lock** connector similar to common handheld appliance fittings. The head is inserted into the circular base and rotated slightly so the locating pins lock into the curved slots, allowing quick replacement while maintaining stable airflow alignment.

Concept Model

Purpose

To demonstrate the final model with features mentioned in a detailed way.

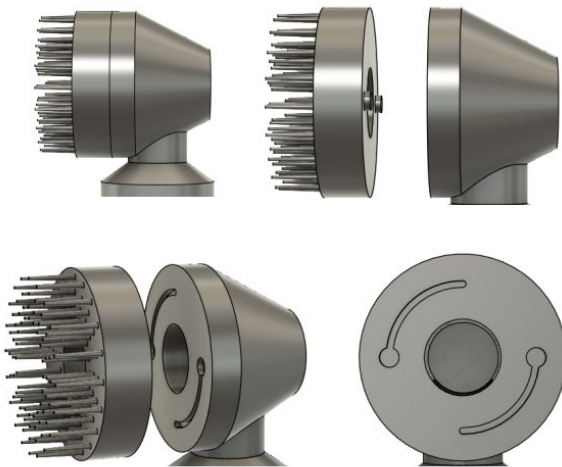
1

Product Overview



2

Head Replacement



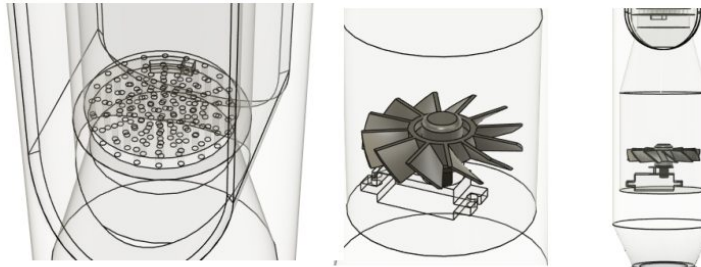
3

Loose Fur Storage Removal



2

Electric Motor & Fans



4

Passive Air Filter (Active Carbon)

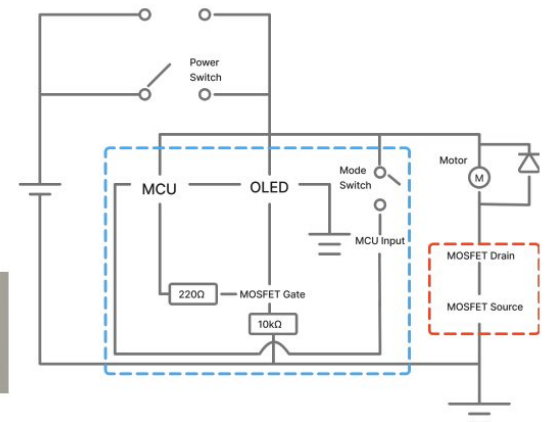


The **electric motor** drives the main internal mechanism of the device, allowing **automated movement** and improving **cleaning efficiency**. The **fan** supports airflow inside the product, helping with **ventilation** and reducing **odour** build-up. Together, they enhance the performance, hygiene, and usability of the design.

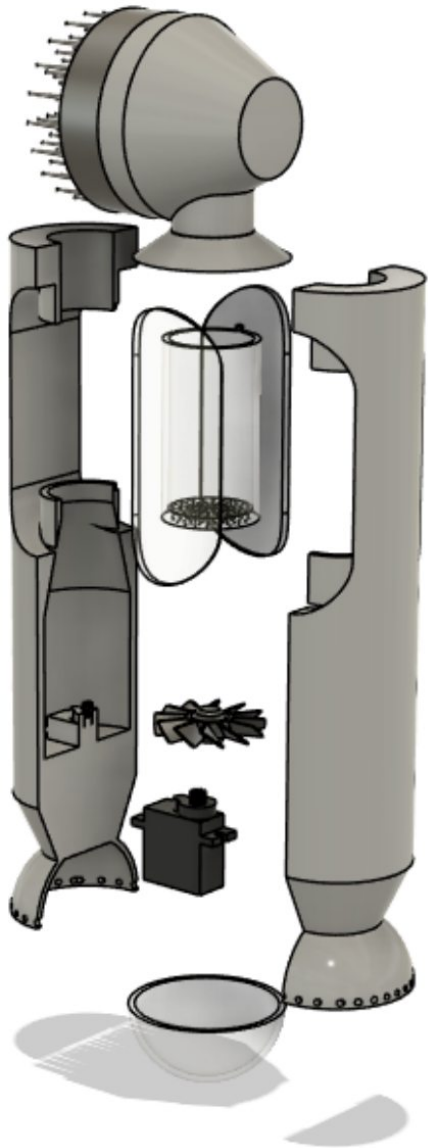
3

Electric Interaction & Circuit

The interface is designed for **simple** and **intuitive** control. A dedicated **power button** switches the device on and off, while a **sliding switch** allows the user to select between low and high fan-speed modes for different debris types and **noise levels**. A small **OLED** display provides clear feedback on battery level and operating mode during use.



Final Concept



Cover	01
Individual Concept & Further Research	02
Initial Ideation	03
Concept Selection	04
Concept Development 1	05
Concept Development 2	06
Final Concept	07

Pure Picker:

Efficient faces cleaning with Better Experience

Individual Work : Cici Song

CID: 02563502

Individual Insights & Further Research

02

Purpose

To derive a clear and detailed design direction from the selected topic of cleaning.

This project aims to identify solutions that **reduce pressure** on **foster care volunteers** supporting **domestic cats** and **dogs**. Three key insights, capturing, feeding, and cleaning, were identified, and **cleaning** was shortlisted as the primary focus, because it offers greater potential for diverse design opportunities, and it is a core task carried out by staff on a daily basis.



1 Insight: Shelter Cleaning

Cleaning and disinfection constitute the **majority** of daily work in shelters, yet outcomes are **inconsistent** due to **variable** hygiene standards and aversion, **driving rework**. **Cross-infection risk** remains difficult to mitigate, while **pet-safe** constraints and **noisy** single-purpose tools restrict effective cleaning in confined, stress-sensitive environments.

2 Opportunity

Design a **quiet, electric-powered** handheld device that automatically collects pet **faeces or pees** into a bag and seals it securely, eliminating direct contact for staff and volunteers. This aims to improve **cleaning efficiency** and **user experience** while **reducing** the **risk** of cross-infection between cats and dogs.

HMW

How might we design a **quiet, pet-safe**, battery-powered **handheld** cleaning device for **shelters** that **automatically cleans up** and **isolate** pet **faeces or pees** without direct contact, so shelter staff and volunteers can clean **faster, reduce rework**, and **lower cross-infection risk** between cats and dogs?

3 Relevent Products

Poop Scooper [1][2][3]



Advantages

- One-hand operation, leaving the other hand free to handle a pet.
- Simple mechanical structure, making it low-cost and easy to use.
- Lightweight and portable, suitable for quick waste collection.

Disadvantages

- Claw or scoop pickup may leave residue and be harder to clean.
- Manual button operation still requires effort.
- Bags must be fitted manually, reducing efficiency for large-scale cleaning.

Pee Management [4][5]



Pee powder absorbs pet urine quickly, reducing direct contact and improving hygiene during cleaning.



Beldray Antibac Pet Plus Spray Mop is designed to clean wet and dry pet messes from floors while reducing the need for direct hand contact.

Sealers [6][7]



Heat Sealer



Twisting Sealer

Initial Ideation

03

Purpose

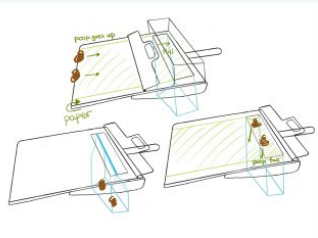
To explore possible solutionst, focusing on pet-friendliness, automation, and user experience.

- Integrated Design
- Individual Stages

Ideas are developed across the full process of collecting cat or dog faeces, placing it into a bag, and sealing it, including both concepts for the **8 complete process** and **7 ideas** for **individual stages**.

1 Poop Picker 1

inspired by pizza peel



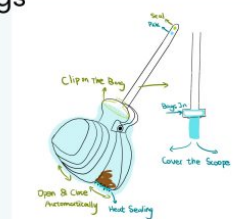
2 Poop Picker 2

use mechanical claw to pick



3 Poop Picker 3

Another claw combing bags



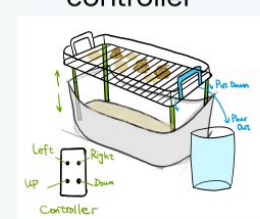
4 Multi-purpose Robot

Cleaning robot with controller



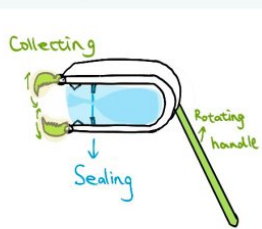
5 Automatic litter tray

Cleaning robot with controller



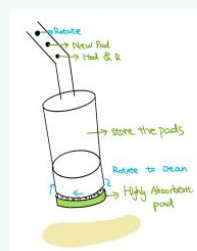
6 Poop Picker 4

A cavity inside to store



7 Pee Picker 1

Absorbent pad cleans up pee



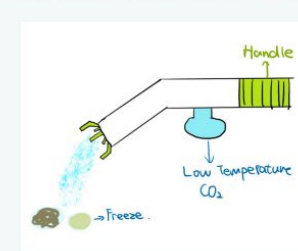
8 Pee Picker 2

Collect pee using a roller



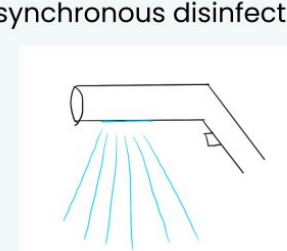
9 Pee Picker 3

Use cold CO2 to freeze



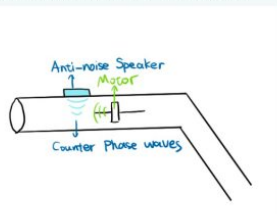
10 Bacterial disinfection

Bottom ultraviolet synchronous disinfection



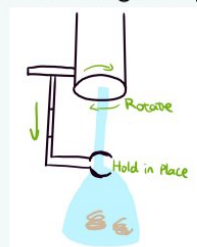
11 Anti-noise Speaker

emits processed counter-phase sound waves.



12 Twist Sealing

Seal the bag temporarily



13 Pee mop

Inspired by whiteboard eraser



14 Detergent Holder

Helpful with odour and pees



15 Heat Sealing

Effective for PE bags



Purpose

To derive a clear and detailed design direction from the selected topic of cleaning.

1 User Review



Shelter Volunteer

"The enclosed cavity in idea 6 makes a big difference for me. Once the waste goes inside, it just feels a lot cleaner than carrying it around on an open tool." --Hemaozhu

"My job would be so much easier if one machine could handle both poop and pee, which is why I like ideas 6 and 7. But honestly, a poop picker feels more needed day to day." -- Ami Cat Shelter

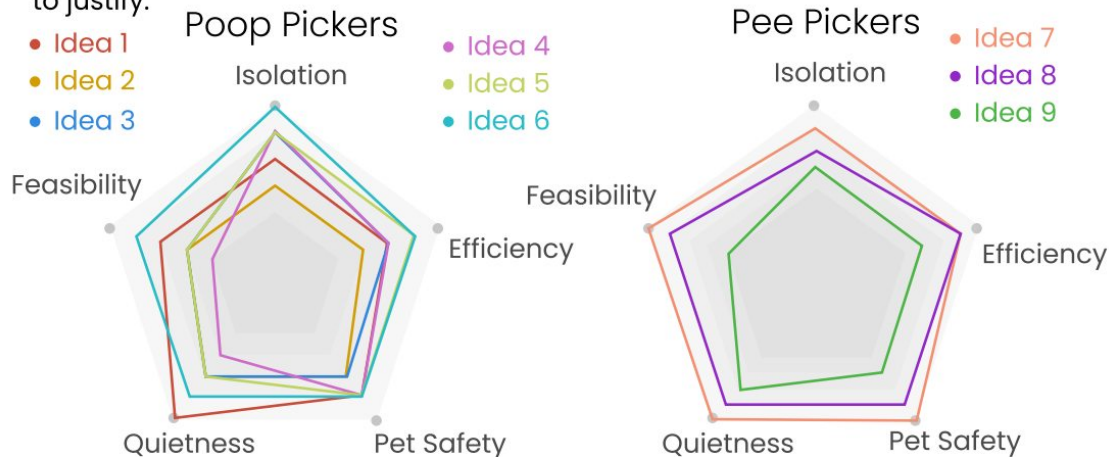
Pet Owner

"My cat does get soft poop sometimes, so if idea 1 works properly, I'd probably trust that more. But otherwise, I think idea 6 is better for picking up poop."-- Leo, owns a cat & a dog



2 Quantitative Rating

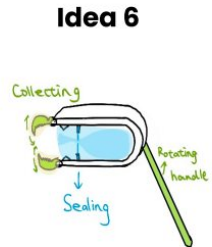
The radar chart helps compare integrated concepts across key design criteria in a clear and balanced way, making the selection process more structured and easier to justify.



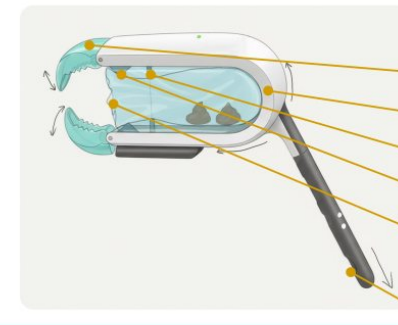
3 Selected Concept

Idea 6 was selected as the final poop-cleaning direction.

- User feedback showed that poop cleaning is a more serious issue that idea 6 feels cleaner, more contained, and less unpleasant to use because of its enclosed cavity.
- It achieved the strongest overall balance in the radar chart, especially in isolation, efficiency, and practicality for shelter cleaning.



4 Final Concept



Poop Picker

- Collection Scoop
- Strage Chamber
- Sealer
- Bag Clip
- Disposable Bags
- Control button

Key mechanisms to build

- Open-close system of the collection flap
 - How can a motor drive the opening and closing movement?
 - Where should the motor be placed?
- Heat-sealing system and protection
 - Where to place the sealer, as the wall of the chamber is not sufficiently thick for full integration?
 - How can a motorised mechanism apply heat for sealing?
 - Where should thermal and hygiene protection be added?
- Internal Structure
 - How to arrange the wires and where to place the main control?

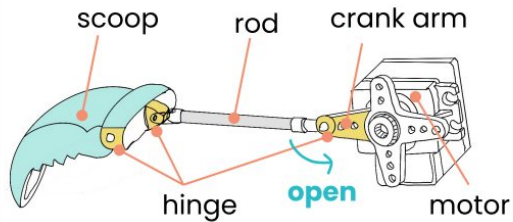
Concept Development 1: Techinques & Circuit

05

Purpose To develop clear mechanisms and circuit systems within the concept to achieve functional goals.

1 Key Techniques

Open-close of Scoop



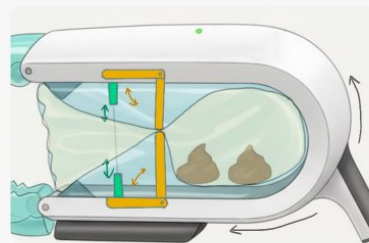
The flap is operated by a motor-driven crank-and-linkage mechanism, which converts motor rotation into controlled opening and closing about a fixed hinge.

- The flap is mounted on a hinge that defines its rotational movement.
- A motor drives a crank arm connected to the linkage rod.
- The rod transfers this motion to the flap near the hinge.

As a result, the flap rotates open or closed in a controlled and repeatable way.

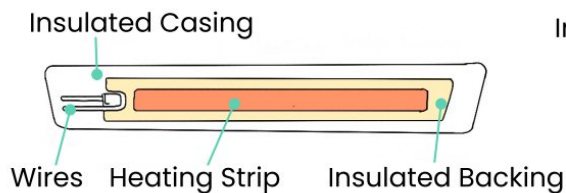
Heat sealing method: folding jaws

The heat-sealing method was changed from a vertical sliding system to a folding jaw mechanism due to the limited thickness of the chamber wall. This allowed the sealing parts to be stored more compactly, while still enabling the bag opening to be clamped and heat-sealed effectively under pressure.

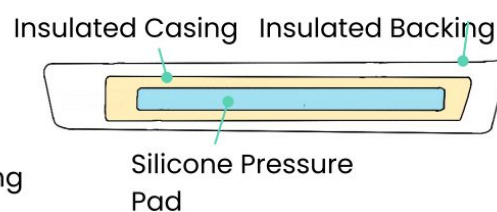


Heat sealer jaws: internal structure

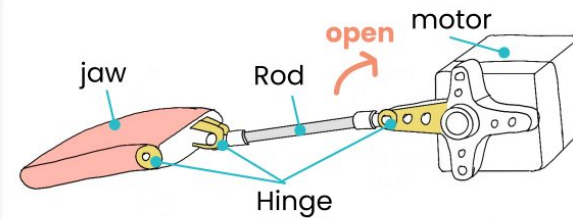
Heating Jaw



Passive Pressure Jaw



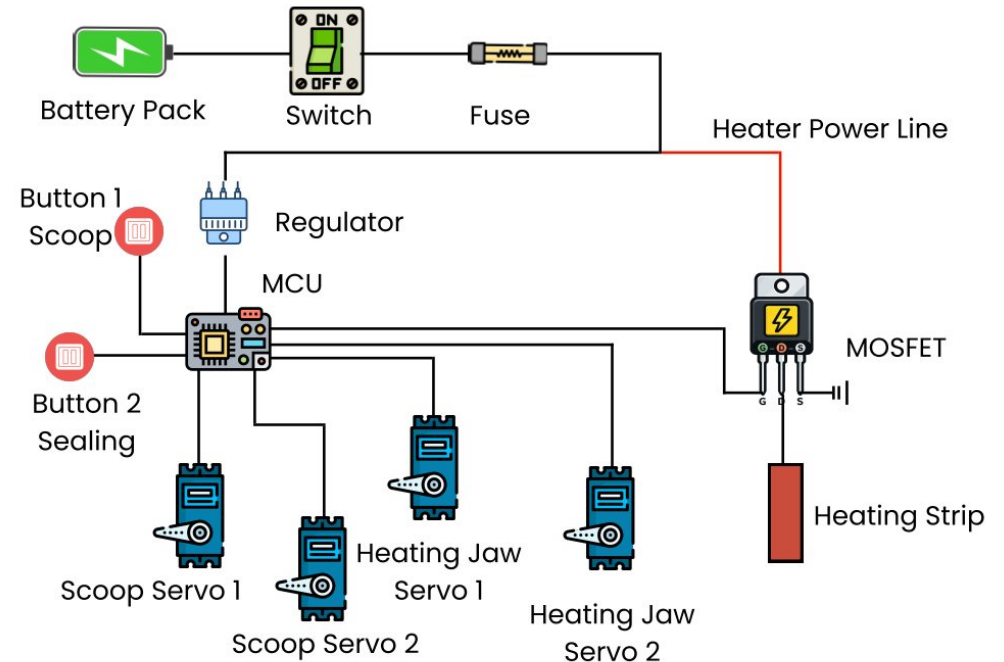
Heat sealing: folding mechanism



The folding jaw is operated through a motor-driven crank-and-linkage mechanism, which converts rotary motor motion into controlled opening and folding about a fixed hinge.

- The jaw rotates around a hinge that defines its motion.
- A motor drives a crank arm connected to the jaw through a rod.
- As the motor rotates, the rod pushes or pulls the jaw.
- This causes the jaw to open or fold back in a controlled and repeatable way.

2 Circuit Structure

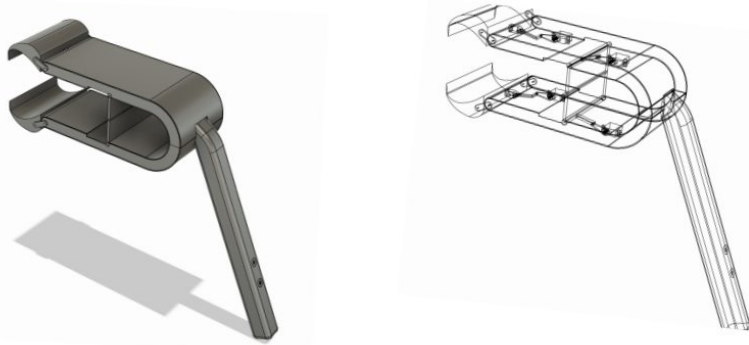


Concept Development 2: 3D modeling & Assembly 06

Purpose To show the mechanism and explain how to build the mechanisms

1 3D Model: Poop Picker

The final concept was modelled in Fusion 360, clearly demonstrating the mechanism of each feature.



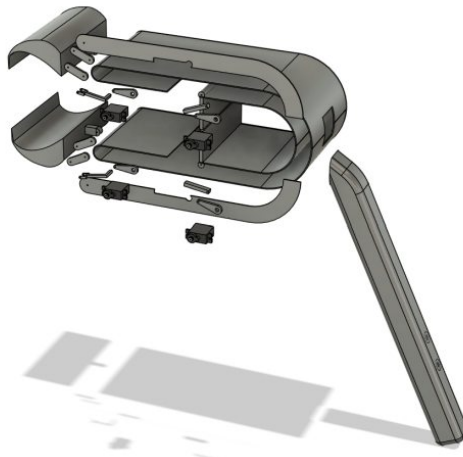
Key Dimensions

- Size of the main body: 500mm * 240mm
- Thickness of cavity wall: 40mm
- Length of handle: 730mm
- Length of the scoop: 137mm
- Max thickness of scoop: 45mm
- Size of jaws: 250mm * 80mm

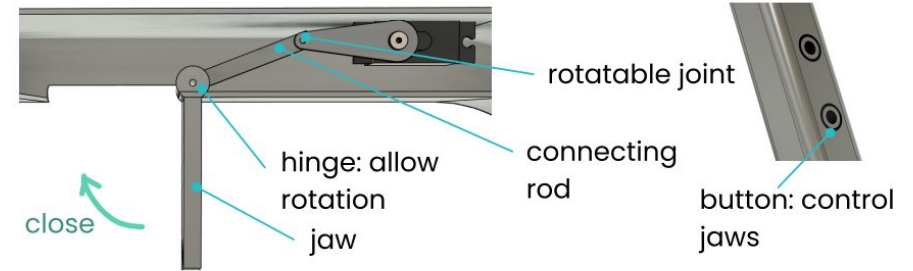
2 Assembly: Exploding Map

The design consists of 43 components in total.

- 3 snap-fit features are used at the handle-to-body connection and cover to support efficient assembly and disassembly.
- 4 screws create rigid connections between the rods and the tabs on the hinges or motor arms, ensuring secure motion transfer and structural stability.



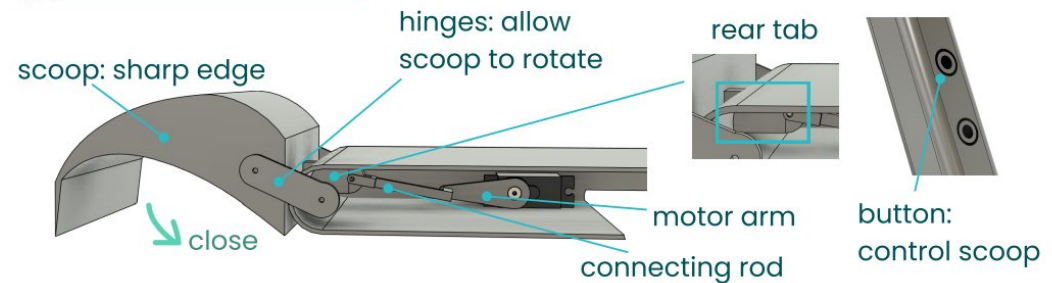
3 Heat Sealing Jaws



The heating jaws use a mechanism similar to the scoop. The motor is mounted on the upper wall of the internal structure to allow space for the motor arm to rotate. The motor arm is connected to the rod by a pin joint, while the other end is fixed to the jaw hinge to transmit motion directly.

The motor arm starts in a left-facing horizontal position. When it rotates clockwise, the rod moves downward and closes the jaw. When it returns, the rod moves upward, allowing the jaw to open.

4 Scoop



The scoop is actuated by a motor-driven linkage. The joint between the motor arm and rod is rotatable, while the rod is rigidly fixed to the rear tab.

The motor arm starts in a left-facing horizontal position. As it rotates clockwise, the rod lifts the rear tab and closes the scoop. When it returns, the tab lowers and the scoop reopens.

Better experience &
faster cleaning
with

Pure Picker



RePooPress



waste management

In Home Environments

Individual Work : Preme Chaisen

Chosen Product Opportunity

HMW design an engineerable solution to manage pet waste in multi-dog home environments while reducing labour and resource demands on caregivers?

CID: 02573426

1 Waste Management : A Crappy Problem

Managing pet waste is an **unglamorous** and **repetitive** part of pet care. Caregivers constantly **pick up faeces, wipe urine** and **vacuum shed** fur to maintain a clean environments. Yet once this waste is sealed in plastic and thrown away, its **environmental impact is rarely considered**.

The Problem Is Piling Up



1.6 Million Tonnes [1]

Dog poop thrown away annually in the UK



124.1 kg [2]

of poo per dog in the UK yearly that's a baby elephant!

1 It's not natural ...

Dog faeces can contain **transmissible pathogens and parasites** (e.g., Toxocara) that contaminate soil, water and other surfaces so it cannot be left behind as fertiliser.

2 We breath it in

10 to 50 percent of the air-born bacteria came from dog poop. [3]

3 It enters our waters [4]

20 to 30 percent of the bacteria in water samples from urban watersheds are from dog waste.



2 Scenario: Home Environment

Target Group

The main focus is on **fosterers or individuals caring for multiple dogs in a home environment**, where one person is responsible for several animals. This includes cleaning the home and yard, as well as managing waste during walks in public areas.

Persona

Dedicated Darcy

52 years old
Experienced Fosterer



Needs:

Standardised solutions that can **deal with multiple animals** and used as a shared resource. Can service multiple dogs before it has to be recharged

3 Key Insights Development

3.1 Interview Insights

Some people have an **innate psychological aversion** to animal waste and are **reluctant** to come in **textural contact** with faeces.

Faeces, expired kibble and fur can be composed alongside enzymes and food-waste to **produce compost**. However, faeces must be air-dried first since they release gases that can kill plants.

Standard **vacuums devices are often ineffective** as pet fur and fluids cling to furniture and surfaces, **promoting bacterial growth**. So, animal-safe solutions such as white vinegar and specialised cleaning sprays are used.

3.2 Deeper Research Pain-points

market insight

Most existing **poop scoopers** are designed for solid pet waste and are **ineffective for handling liquid waste** such as diarrhoea or **urine**. Especially, **during walks** where appropriate **cleaning tools are not readily available**. As a result, **waste** is often left on pavements or in parks, **creating public health and hygiene concerns**.



waste management

Animal poop **contains pathogens and parasites** that can infect humans, so **safe composting methods** must be used, which can be **difficult for beginners** to apply.

Insights:

- Biodegradable plastic bags degrades slowly, making it ineffective for composting. [5]
- Most cat litter are non-degradable
- Compost cannot be used for eatable plants
- Cat poop is more [6] dangerous than dog poop

safety hazard

The cleaned area and the user must be **properly disinfected** to prevent contamination. **Young children** are especially **vulnerable** to roundworms because they often play on the ground and put objects in their mouths.

Potential Risks:

Illness and infections (E. Coli, Salmonellosis)
Blindness (Round Worms)
Abortion (Toxoplasmosis) [7]

3.3 How Might We?

1. Ease & Comfort

HMW **reduce the tactile, sensory, and visual discomfort** of cleaning pet waste while lowering the **labour demands** for people managing **multiple dogs**?



2. Repurposing

HMW streamline the **process of managing and composting** dog or cat waste to **reduce time and labour and animal waste**.

3. Pet-Friendly Sterilisation

HMW enable users to easily **clean contaminated surfaces** caused by pet waste to improve hygiene and safety in **shared environments**?

Concept Exploration

HMW

Ease & Comfort

Repurpose

Sterilise

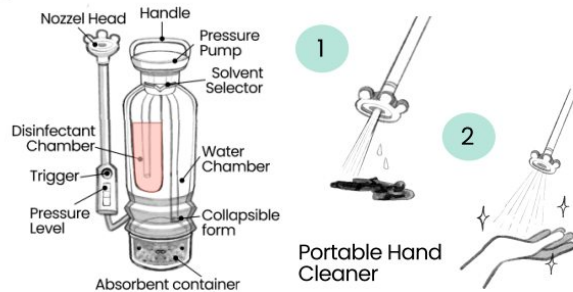
P ass

02

Purpose

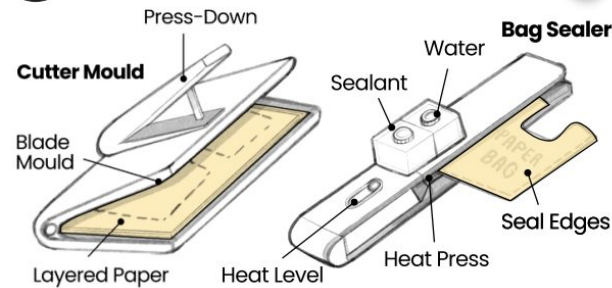
To explore **diverse design concepts** for **pet waste management** in a home environments **base on our HMWs**

1 Portable Water Pressure Spray



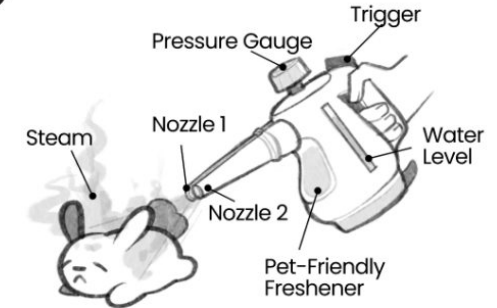
Device for cleaning pet waste that **cannot be picked up with a bag**. Absorbent pellets are first poured onto liquid waste to **clump** for easier collection. Remaining residue is then **sprayed away** while a separate chamber sprays **disinfectant to sanitise** the surface. Suitable for pavements and other hard surfaces **during walks**.

2 Paper Poo Bag Maker



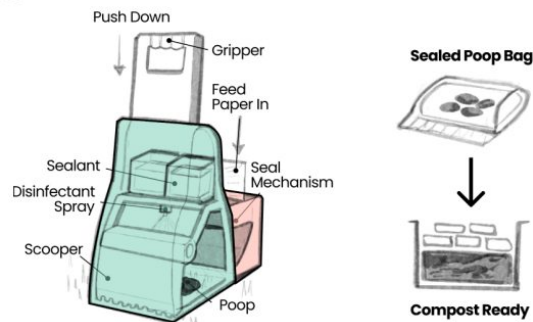
Device that **recycles scrap paper into durable, compostable poop bags** to decrease the use of plastic bags. The sealant and absorbance layers **prevent moisture from animal waste from seeping** through to outside contact. Opportunity to add odour preventive measures.

3 Gentle Pet Steam Cleaner



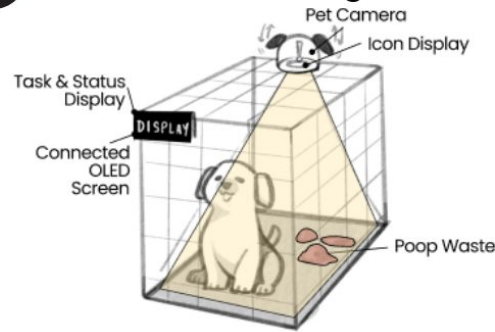
Cats can develop **dried faecal matter stuck in their fur** that is difficult to remove. This handheld device releases **gentle warm steam** to **soften the debris**, allowing **quick and hygienic removal** with a wipe. It can also be used for spot cleaning dried mud and light cleaning between baths.

4 Multi-Poo Scooper & Preper



Area multi-poop scooper with built-in news **paper sealant mechanism**. Sealed compact poop bag can be **directly deposited in compost or wormeries** unlike plastic bags. Automatic **sprays disinfectant on contaminated area**.

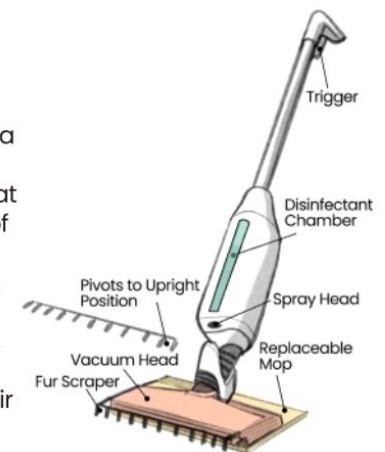
5 Pet-task Monitoring Camera



Device **attaches to kennel** and monitors that area. It **tracks condition of area** and **alerts caretakers** and volunteers if assistance is needed. Connected screen **displays recorded completed task** and **frequency**, flagging any **potential anomalies**.

6 3 in 1 Pet-Orientated Cleaner

A device that **vacuums and mops floors** to remove pet mess. The mop includes a **replaceable absorbent pad** that can be disposed of after soaking up urine. A **front rake, inspired by fur-removal brushes**, lifts and collects embedded pet hair from carpets.



Concept Selection

Purpose

To **reflect** on user **feedback** and **select an concept**.

03

1 User Feedback

Tim (Dog Owner & Walker)



During Walks:

“We use plastic bags because I **want the guarantee** that when I grab the poop, it won't dirty my hands.”

Critical Feedback

- For a portable device to be adopted on walks, it has to be **small enough** so that it can be **grab and go**, either holdable or attachable to the hip.
- People don't carry and prepare a designated pet bag.

Ranked Concept

- 3 Gentle Pet Steam Cleaner
- 2 Paper Poo Bag Maker
- 1 Portable Water Pressure Spray
- 4 Multi-Poo Scooper & Preper

2 Team Assessment

✗ Rejected Concepts:

- 1 Difficult to **miniaturise** a water pressure spray to a portable, bottle-sized form.
- 3 Requires direct **animal testing** to evaluate effectiveness.
- 4 **Overcomplicated**, similar outcomes can be achieved without a battery-powered solution.

“Since it's an **outdoor tool** it seems unnecessary to add battery power component (Team Member)”

2 Paper Poo Bag Maker

Ranked highest consistently amongst the feedbacks.

Project Constraints:

Hand operated battery-powered device with a **rechargeable battery**.

Many ideas suit mains power, so this constraint means:

- Operates in short bursts (not continuous)
- Battery lasts for a full interaction
- Clear justification for why electronics are essential to the function

Fatima (Fosterer)



Favourite Idea?

2 Paper Poo Bag Maker

“I really like the second idea. The **others seem like it would cost a lot**” (*financial strain*)
“The **recycle and reuse** concept seems helpful” (*repurposing*)

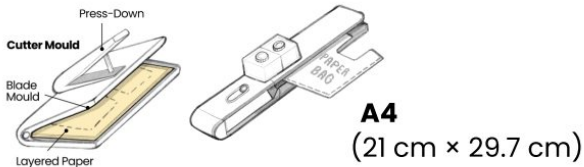
Why would this be helpful?

Foster and shelter carers often care for **multiple dogs** with **limited financial support**, making them prioritising essentials like food and medical care over additional costs.

Using scrap paper as a low-cost alternative to poop bags could help **save up to 1,000 bags per dog each year**. By reducing cost and increasing accessibility, it may **encourage** more **consistent waste pick-up** and **promote responsible disposal methods** like composting instead of landfill.

Selected Concept

2 Paper Poo Bag Maker



This concept was chosen as it offers a **low-cost alternative to plastic bags** while addressing the **need for alternate disposal method** which could then help lower the environmental impact of owning pets.

3 Key Product Specification

Low-cost & Accessible

Foster carers often face financial constraints, so using **readily available materials** like scrap paper **reduces ongoing costs** and lowers the barrier to consistent waste management.

Efficient A4-to-bag

Caregivers manage multiple dogs and repetitive tasks, **so a fast, simple process** reduces effort and encourages regular use.

Durable & Compostable

Bags must **reliably** contain waste while supporting environmentally **responsible disposal**, helping reduce reliance on plastic and landfill impact.

Development 1

Purpose

To **redefine the problem** and **develop the paper poo bag maker**

04

1 Redefined: Problem Statement

Dogs caretakers store dog poop in small plastic bags. That's **1000 plastic bags per dog** annually! [8]



The Current Solution

There are 2 popular options:

Plastic Bags



Lightly scented with mint [9]

Biodegradable Plastic Bags



[10]

The Toxic Green Lie



Biodegradable bags are actually **worse for the environment**. They are designed to be composted, however most of them **end up in landfills**. As they degrade, they can **release untreated waste**, spreading **contamination and disease**. [11]

Not Compost Ready?

Biodegradable plastic poops bags marketed as compostable **do not degrade fast enough** for composting, especially in a **home-composting set-up**. Some [12] composting facilities separate the waste from bags before processing.



[13]

2 Proposed Solution

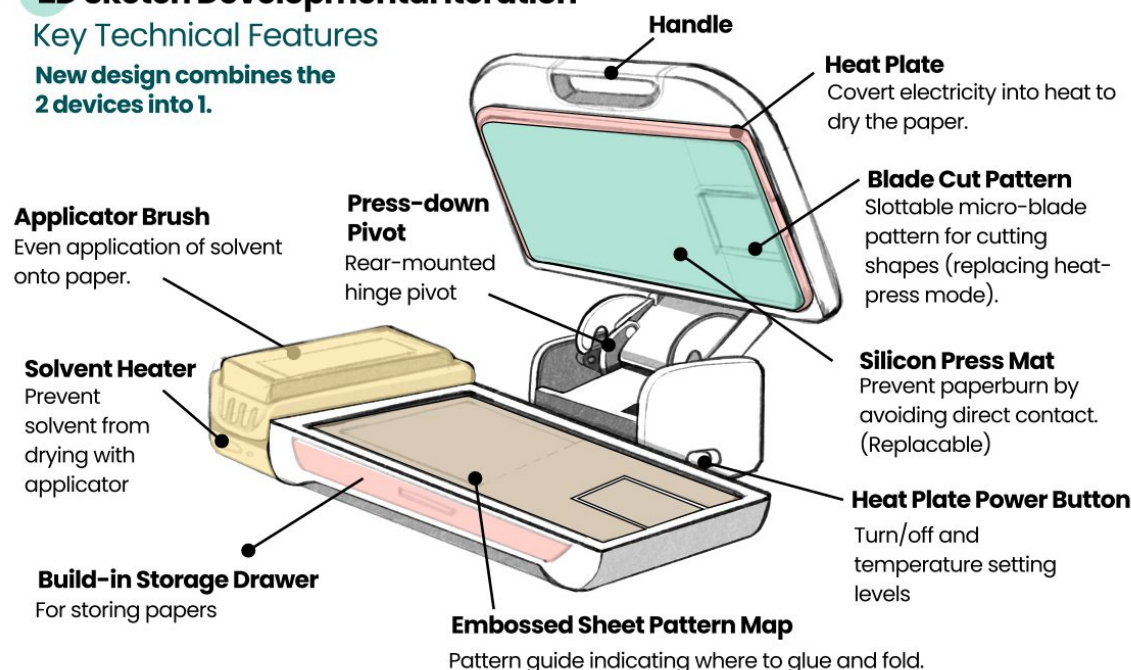
Fosterer insight: Many UK shelters require fosters' to **have a garden**, making home composting feasible.

The solution is to design a device **that converts scrap paper into waterproof, compostable paper bags** that **match the performance of plastic poop bags**. The caveat is that it **assumes** the bags are **disposed** of in **composters** or **wormeries** at end of life.

2D Sketch Developmental Iteration

Key Technical Features

New design combines the 2 devices into 1.



3 Viability Check and User Interaction

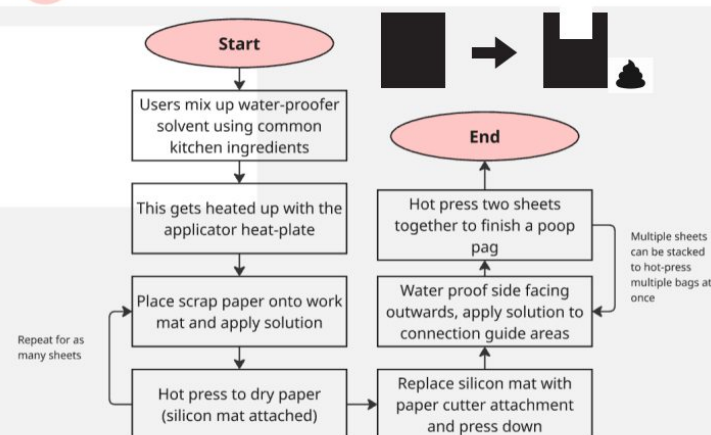
Waterproofing Paper?

Faux plastic can be made from common kitchen ingredients by **applying** this solution **to paper** and **heat pressing** to dry and smooth it. It also acts as a **natural adhesive**.

Biodegradable Waterproof Solvent Recipe [14]

- 1 cup water
- 1 tbsp cornstarch (mix thoroughly)
- 1 tsp vinegar (preservative to prevent bacterial growth)
- 1 tsp coconut oil (improves flexibility)

3.2 User-Interaction Flowchart



There are **7 steps**, which can become complex. The design must use **clear affordances** to show each stage, enabling users to **complete actions in single, intuitive motions** (e.g. brush across, press down).



RePooPress

Re-pur-pose scrap paper into
waterproof pet poop bags

Perfect for **home-composting** and **wormeries**



Pick



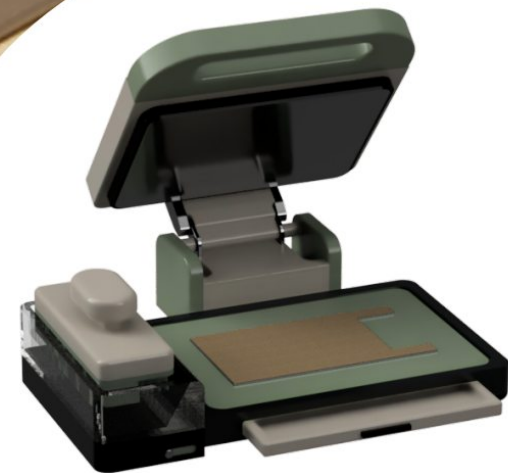
Invert



Go

Why paper poop bags?

Current pet waste disposal relies heavily **on single-use plastic “biodegradable bags”**. When in landfills, these degradable bags can cause the spread of harmful pathogens and parasites while not degrading fast enough for composting. By rethinking how waste is contained at the point of collection, this project aims to **reduce plastic dependence** and **promote** safer, more conscious **waste management habits** like **composting** and **wormeries**, especially in homes.



Fusion360 Render

Dermascan Grooming

For the less experienced

Individual Work : Sirin Vivatanaprasert
CID: 02613317

Cover	01
Concept rationale and Insights	02
Concept Exploration	03
Concept Selection	04
Concept Development 1	05
Concept Development 2	06
Final Concept	07

Concept Rationale and Insights Grooming

02

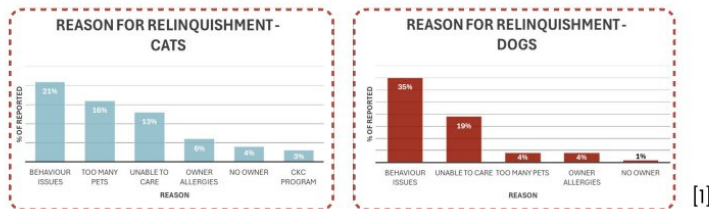
Framing an **individual HMW**, further **insight development**, and assessing **project technical constraints and other specifications**

Key Insights + Design Opportunities

Individual HMW

Secondary research

- 1 Shelter websites often **do not require formal training** for prospective volunteers to handle animals, which can lead to **limited experience**, resulting in **incomplete grooming** and a **higher risk of handling-related accidents** →
How might we make grooming more precise without relying on the skill of the carer?
- 2 The **number one cause of animal relinquishment in the UK** is **behavioural issues**. Animals usually come into shelters **stressed**, and **aggressive**, as such making **check ups** and **grooming** activities more **difficult** for volunteers →
How might we make check ups and grooming tasks that require direct contact with animals safer for volunteers?



User Insights

- 1 **Fur management** is one of the most **repetitive tasks** for pet owners. Dogs often need **daily blow-drying** to remove loose fur and **reduce shedding**. While this is **typically done outdoors**, it is **not always possible in indoor shelters**, leading to a build up of loose fur and mess inside →
How might we manage and reduce loose fur in indoor shelters in a more clean and efficient way?
- 2 **Domestic animals**, especially **cats**, are often very good at **hiding signs of illness**. One pet owner we interviewed reported **recurring skin issues**, such as ringworm or fungal infections, even while the animal was receiving treatment →
How might we help carers identify hidden skin issues during grooming, especially in animals that show few visible signs of illness?

Current Market

- **Limited products** that assist users in **diagnosing skin issues**, with focus mainly on medication after diagnosis
- Many products operate under the assumption that **animals are willing for their carers to be in contact with them** and often don't consider **aggressive animals**



Grooming is a **time consuming** and **repetitive** task in animal shelters that requires **volunteers** to work in **close contact** with cats and dogs. As **many animals may be stressed or unpredictable**, this can create **safety risks**, particularly for **less experienced volunteers**. This highlights the need for solutions that make grooming **safer** and **easier** for this user group.

HMW make grooming experience for less experienced carers and volunteers in animal (cat + dog) shelters safer, easier to perform, and more precise?

Technical Constraints

- 1 **Battery powered** (No power cords)
 - Maximum Voltage required to power device should not exceed **12V**
- 2 **Handheld device**
 - Design should have a maximum weight of no more than **2kg** if designed to be **portable**
 - If designed to be **fixed**, **longest length** should not exceed **1m**

Other Requirements

Affordable

Product should not run on a single use → dispose model, and should not exceed the average market price of a similar item

Safe

Product should not endanger the handler or animal, and should have embedded safeguards in the case of mishandling

Simple

Product should be relatively self explanatory and new handlers should not have trouble operating the device

Durable

Product should be able to withstand constant usage, last for at least 1 year, and have repairable components

Concept Ideation 20 Initial Ideas → 9 Detailed Sketches

03

Coming up with 20 initial ideas and selecting 9 for detailed sketching based on prototyping feasibility and specifications in P.2

20 Initial Ideas



- Indirect assistance – support processes
- Assistance with areas that are difficult to access
- Direct grooming tasks

1 Vacuum hair compressor

Cyclone bundles fur together

Moves **down** to act as **hair compressor** and **up** to reset position

Temporary fur **storage**

Opens up to let fur fall down

Compresses large amounts of fur in vacuum cleaners to a flattened pad to **reduce clogging and disposal frequency.**

2 Smart leash

Motion and Heartbeat sensor on collar senses a **pounce** and **tightens leash**

Leash **extends** for **free roam** (return some freedom for the dog) and **retracts** for **scheduled grooming events**

Use a **semi-restrictive leash** to safely manage potentially aggressive dogs, with **scheduled retraction** to prepare them to go for grooming, reducing **handling effort** to catch the dog.

3 Clothing fur roller

(Electrostatic)

MOTOR

Motorised spinning roller

Vacuum sucks hair from roller to keep roller clean

Traditional **lint rollers** fill up quickly, and **vacuums** can be **harsh on clothing**, so a motorised hybrid system offers **longer use** before having to dispose of fur and is **gentler on fabric.**

4 Skin issue checker

Live footage on screen

Camera captures footage of skin

Parts hair blocking vision of skin and **projects live footage** on a separate screen – (addition) A **machine learning program** can also be written to **automatically flag issues found** during skin sweep.

Vacuum nozzles part fur to reveal skin underneath

5 Cat underbody check toy

(carer can adjust to specific toy and height)

Toy moves along tracks

Exposed underbody for carers to visually check

Bottom half of installation is **see through**

Helps guide cats into a **position** where carers can see their **underbody** by relying on their tendency to pounce towards moving toys. This also doubles as a **play wall**

6 Teeth checker bone

Bite strength and inside mouth **video check**

IMAGE **BITE STRENGTH**

Scent diffuser

Camera **Bone chew toy**

Helps guide cats into a **position** where carers can see their **underbody** by relying on their tendency to pounce towards moving toys. This also doubles as a **play wall**

7 Vacuum glove

knobs comb through fur

Vacuum takes up loose hair

Bite safe

Combines play time with groom time – when petting animals **knobs** comb through fur and **vacuum** sucks up loose hair. Glove is **bite safe** to increase the safety of volunteers and carers.

8 Nail trim guide

Angled camera and **laser**

Adjustable paw locking mechanism

Nail grinder

Paw is locked in place and **thermal or IR camera** detects where nail should be cut based on distance from skin. A **laser guide** is then shone on the nail, where a **nail grinder** can be used.

9 Paw washer

Blow dryer

Water spray

Pressure sensing pad

Targeted at animals carrying **dirt from outdoors**. **Pressure sensor** detects animal presence before **spraying** their paw with **water** and **blow drying quickly**

Concept selection

User Input and Prototyping feasibility

04

Selection of a final concept based on feasibility, and 3rd party inputs from 4 users, and a product designer

Early idea elimination

The **smart leash** (2) and **nail trimming guide** (8) were **eliminated** due to **ethical concerns about restraining animals** - this was later **confirmed by users** who mentioned the product "is **too risky** and might hurt the animal if they move around". The **teeth checker** (6) was also removed, as shared use could increase the risk of **disease transmission** between animals.



User ranking of ideas (best = 1)

User	1	3	4	5	7	9
User 1	3	2	1	4	5	6
User 2	3	2	1	-	5	4
User 3	5	2	1	6	4	3
User 4	5	2	1	-	4	4

"Which idea will help assist you in your grooming needs the most, why?"

Overall Ranking

1. Skin issue checker
2. Clothing fur roller
3. Vacuum hair compressor
4. Paw washer
5. Vacuum glove
6. Cat underbody check toy

User feedback

"My favourite is the skin issue checker"

"This feels innovative cause it's easy to miss rashes or ticks and is especially good for long haired dogs"

"Love this! My cats have skin issues very often - scratches, ringworms - where will this be sold?"

"My dog loves to bring back ticks when he goes outside so this would be great for finding them"

"This one is super interesting - would work great with my dog, and I'd imagine vets would want one of these too"

User 2 -
Dog owner

User 3 - Dog +
Cat owner

Product designer -
Ex Dog owner

User 1 - Multiple
cat fosterer

User 4 - Dog
owner



Tips

- A good product will not try to **go against animal nature**
- Consider similar **alternatives** for some of these designs that require **less motor power**

Selected final design



Idea 4 - Skin Issue Detector

Although other designs such as the **clothing fur roller** (3) and **vacuum hair compressor** (1) received positive comments from users such as "roller is nice as I don't have to peel the sticky tabs as much" and "vacuum hair compressor is cool cause I won't have to go to the bin as often", there was an **overwhelming consensus** that the **skin issue detector** (4) was the most innovative and tackled a very common issue that pet owners seem to have, but have assisted solution for.

The paw washer (9) and cat check toy (5) received mixed reviews as users mention it can be hard to get their pets to cooperate.

Further development of idea, design inspiration from existing products, ergonomics, 3D modelling, and user workflow

Developed Design

Multiple vacuum strengths to handle thicker fur

Part lines where parts can be taken apart, either to **dispose fur** or for **maintenance**

General geometry based on **market handheld vacuums**



Ergonomic grip

Based on: Handle of a Thermos of similar geometry



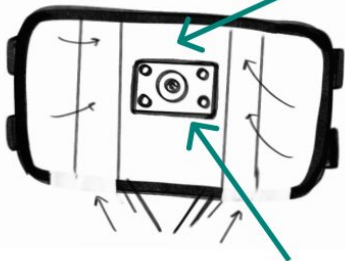
Comb strand parts, de tangles, and **guides** fur to the side



Based on: Human hair parter end

Soft wheels act as a **barrier** preventing the edges and comb strands from **scraping** the animal's skin.

Using a **dark interior** will help **reduce light reflections** and glare from the LEDs, improving image clarity by **minimizing unwanted light scattering**.



Camera with LED

Live footage



A **machine learning model** identifies potential skin **anomalies** by comparing images to a trained dataset.

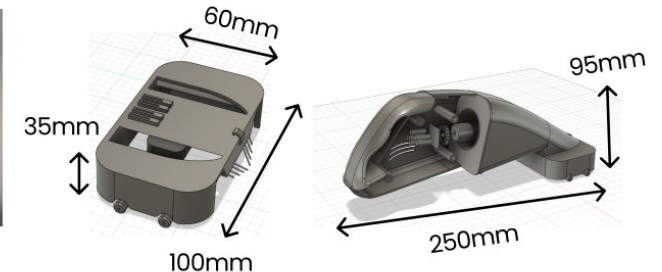
Product Name: Dermascan

A handheld grooming device that parts an animal's fur to expose the skin and uses a camera with an LED to capture live footage. Integrated suction with a low power vacuum helps hold the fur aside and collect loose hair, while a machine learning system analyses the images to flag potential skin issues. The device is designed to assist volunteers in identifying skin problems more easily while carrying out routine grooming tasks.

Dimensions



Approximated using a cardboard box



Usage workflow

1. User syncs camera with an external device to view real time images



2. User powers on the device and glides it across the animal's fur



3. Screen displays image and flags anomalies



Concept Development 2

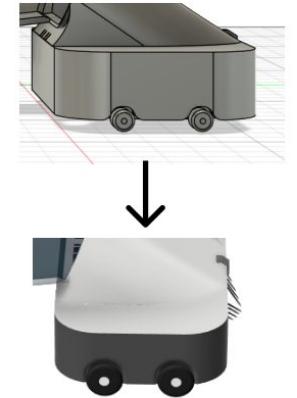
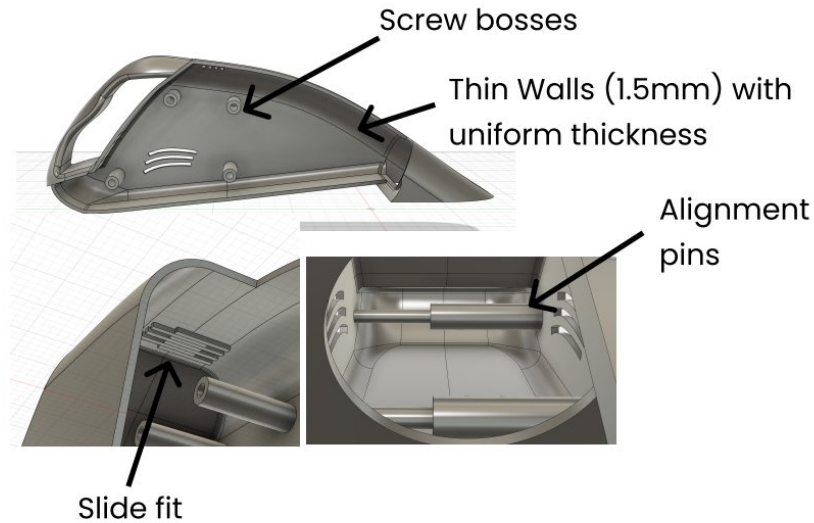
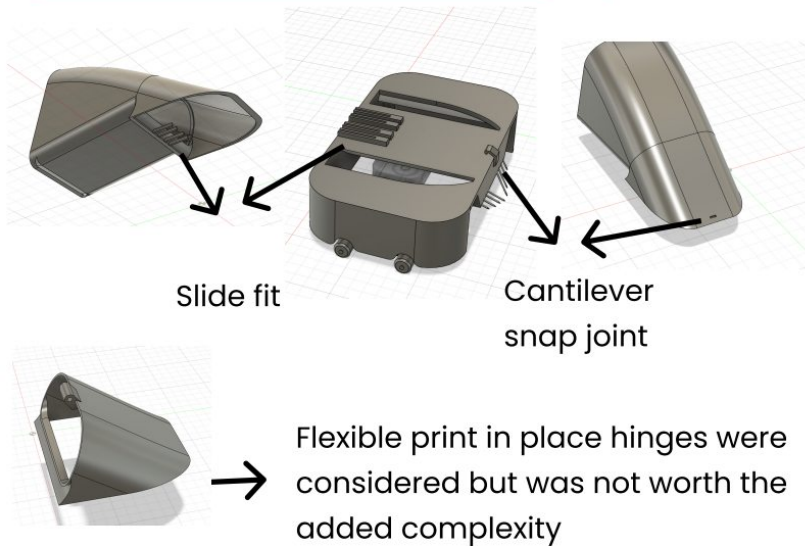
DFA/DFDA and Final Product 06

Final development of 3D model, DFA considerations, key electrical components, and features breakdown

Design for DFA/DFDA

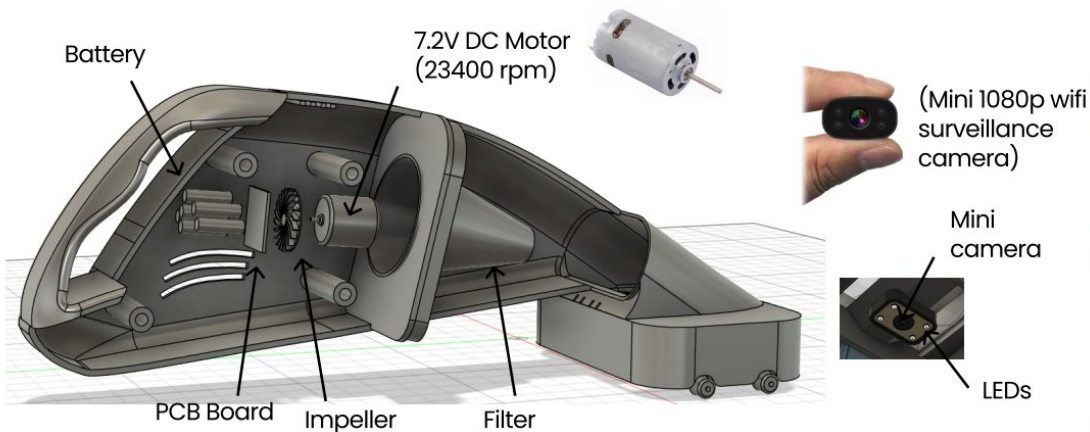
Sliding fits and Snap fits

3D printable design

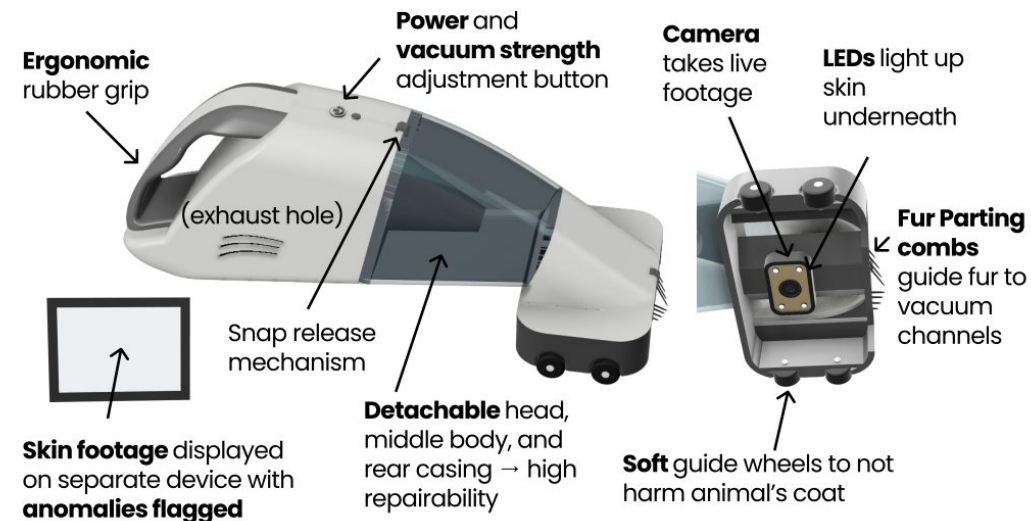


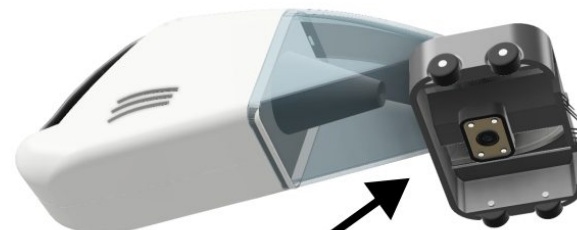
Key Electronic Components

Only the most important components are modelled. For prototyping, an existing hand held vacuum can be taken apart and modified to fit the design.



Features breakdown





(Fusion 360 render)

Scan with the help of the fur partter and live camera + LED

ANOMALY
DETECTED

Detect skin anomalies through footage by comparing with a reference photo library

Provide the necessary **care**

why

Skin conditions are common in cats and dogs, including infections, wounds, and parasites that can cause discomfort if not treated. Early detection is important, as many conditions can worsen over time but are easier to manage when identified early.

However, spotting these issues can be difficult due to an animal's fur, particularly in thicker coats. Less experienced volunteers may also find it challenging to recognise early signs, meaning problems are often identified at a later stage. This product aims to support carers by making it easier to detect potential skin issues during grooming.

Dermascan

Scan. Detect. Care.

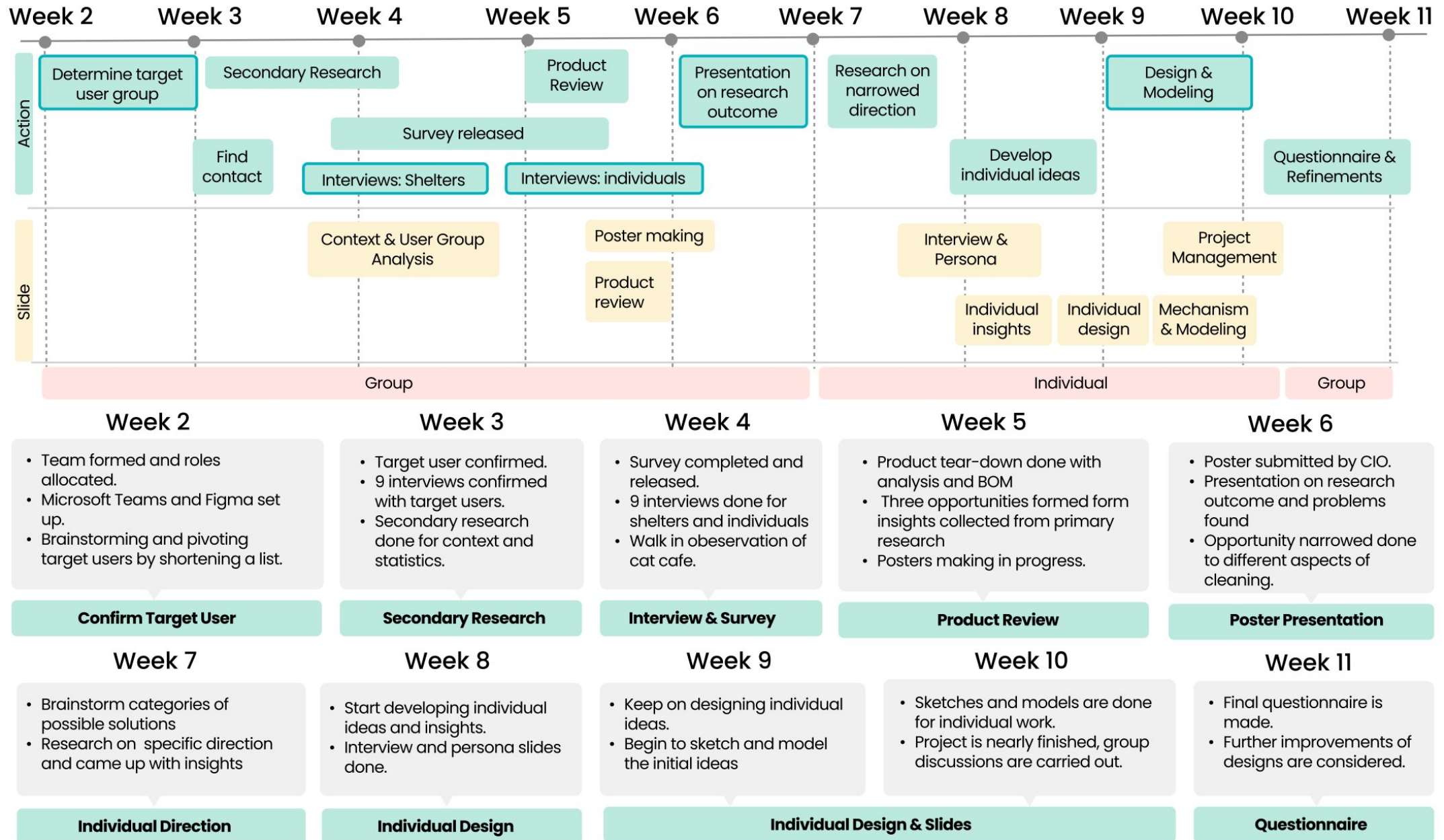
Project Timeline

Purpose

To show the plan and project working timeline.

12

Key: Action Documentation Section



Group Work

Purpose

To demonstrate our groupwork approach and the challenges encountered.

13

1 Teamwork Management

During the project, we hold **regular meetings** once or twice a week to share progress and align on assigned tasks. All group members **actively contribute** in their **respective roles** throughout the design process. **Figma**, **Microsoft Teams**, and **Miro** are the primary tools used to coordinate and manage the work.

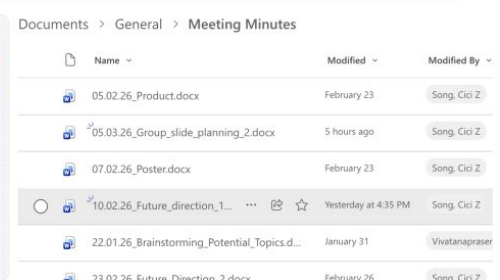
- **Figma** is mainly used to develop slides, including early drafts, as well as posters, and to support brainstorming, idea development, and ongoing research updates.
- **Microsoft Teams** is used to store and organise shared files, including meeting minutes and interview transcripts.
- A **Gantt chart** on the Miro board is used to track tasks, monitor progress, and manage timelines.
- **12 meeting minutes** were created to document each meeting and maintain an up to date record of task progress and next actions.[12]



Documents > General

Name	Modified	Modified By	+ Add column
Consent Forms	January 30	Vivatanaprasert, Muk	
Interview transcripts	January 30	Vivatanaprasert, Muk	
Meeting Minutes	January 30	Vivatanaprasert, Muk	

(link to drive in references[12])



Documents > General > Meeting Minutes

Name	Modified	Modified By
05.02.26_Product.docx	February 23	Song, Cici Z
05.03.26_Group_slide_planning_2.docx	5 hours ago	Song, Cici Z
07.02.26_Postor.docx	February 23	Song, Cici Z
10.02.26_Future_direction_1...	Yesterday at 4:35 PM	Song, Cici Z
22.01.26_Brainstorming_Potential_Topics.d...	January 31	Vivatanaprasert
23.02.26_Future_Direction_2.docx	February 26	Song, Cici Z

Strengths

- **Alignment (12 meetings)**: Weekly 1–2 check-ins with 100% attendance help us agree on goals, priorities, and next actions, reducing misdirection.
- **Coordination**: Team members contribute consistently within their roles and hand off work smoothly across dependencies.
- **Accountability**: Tasks are linked to clear owners and deliverables, making follow-ups and progress checks straightforward.

Future Improvements

- **Micro-deadlines**: We should break tasks into smaller, manageable sections with short checkpoints to improve pacing, visibility, and on-time delivery.
- **Post-interview review**: After each interview, we should have a short discussion on the key insights and clarify any differences in interpretation.

2 Challenges Faced

1 Difficulties in Building Contact

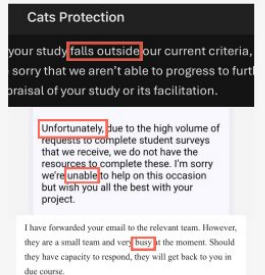
Emails to rescue centres and shelters often received delayed responses or were rejected.

Why

- Limited staff capacity
- Some organisations do not support student research

What we did

- Recruited individual fosters and pet owners
- Conducted a cat café observation
- Completed 9 interviews in total



2 Content Prioritization

Interview data exceeded the page limit.

Why

- 3 sub-divided user groups and 9 interviews in total gives comprehensive perspectives.
- Large amount of quotes and painpoints of each interview.

What we did

- Grouped findings into key themes.
- Kept only the most representative evidence.
- Reduced repetition and improved readability.



3 Project Reflection

Overall, our teamwork **worked really well** and **felt collaborative** throughout the whole project. Regular meetings, clear roles and a structured workflow helped keep the **project organised**. Despite **tight timelines**, we still turned a large amount of interview data into clear themes. **In the future**, micro-deadlines and short debriefs could make the process even better as we look towards a bright **prototyping phase**.



Questionnaire

14

Thank you for reviewing our research and concept booklet. Your feedback will help us evaluate and improve our proposed pet-waste cleaning device concepts for shelter environments. Please complete the short questionnaire below.

Full name:

Which animals do you mainly work with? Cats / Dogs / Both

Date completed: / /

Which best describes your role? (Tick one) Volunteer or staff member in a pet shelter / Individual foster carer

PLEASE RATE THE IDEAS FROM 1–4 (BEST TO WORST)

Dermascan



Pet Ease



Pure Picker



RePooPress



Which concept best improves the efficiency of daily cleaning tasks?
reducing time, repetition, and overall workload in shelter.

Which concept feels safest and least stressful for animals?
considering noise level, movement, and interaction during use.

Which concept seems safest and most hygienic to operate (for people) ?
considering risk of direct contact, containment of waste, and ease of cleaning after use.

Which concept best reduces resource strain?
reducing financial strain, time-consumption and physical stress.

Which concept is most practical for repeated, long-term use in shelters?
considering durability, consistency, and ease of maintenance.

Which concept seems easiest to learn and use quickly?
considering clarity of operation, number of steps, and ease of first-time use.

Which concept seems most suitable for use by volunteers with different experience levels?
allowing both new and experienced users to operate it with confidence.

Which concept would make you less resistant to doing waste-related cleaning tasks?
reducing the discomfort and inconvenience often associated with these jobs.

Which product do you see yourself buying/using?

Reference 1

Group References:

- [1] RSPCA, Animal Kindness Index Report 2023. [Online]. Available: <https://www.rspca.org.uk/whatwedo/latest/animalkindnessindex>
- [2] RSPCA, "Abandonment of animals rises due to cost of living crisis." [Online]. Available: <https://www.rspca.org.uk/-/news-abandonment-rise-cost-of-living>
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